



សាលាបង្រៀនអោយដឹង

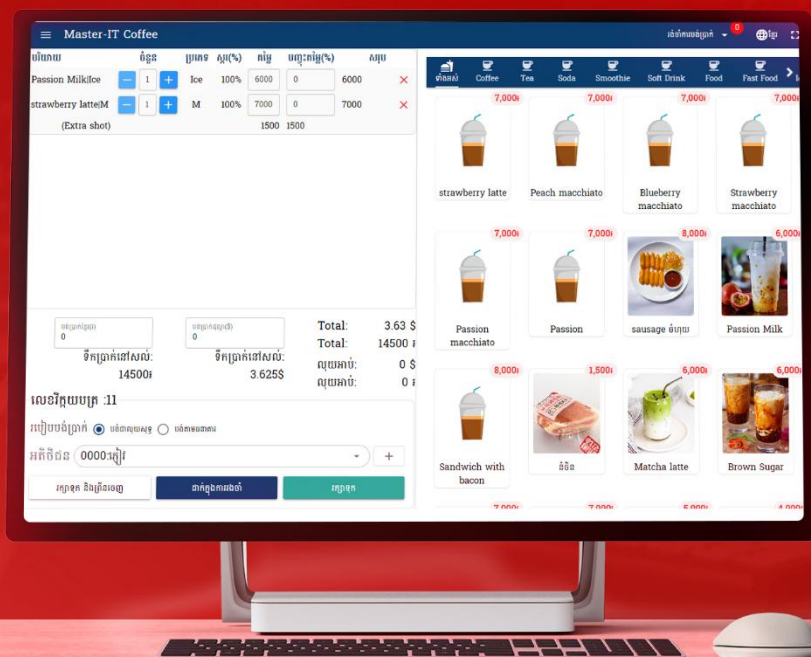
បង្រៀនជំនាញ បង្វឹកអាជីពសម្រាប់ការងារនិងមុខរបរ

ត្រួតពិនិត្យជា

អ្នកជំនាញអភិវឌ្ឍន៍ System

POS System

ជួយឱ្យអ្នកសិក្សាបានបទពិសោធន៍ពីការបង្កើតប្រព័ន្ធគ្រប់គ្រង POS System



2024 - 2025



អារម្ភកថា

ដោយឃើញថាសព្វថ្ងៃនេះបច្ចេកវិទ្យាកំពុងតែការរីកចម្រើនពីមួយថ្ងៃទៅមួយថ្ងៃ ជាពិសេសគឺការប្រើប្រាស់កុំព្យូទ័របានជំនួស និងសម្រួលដល់ការងារមួយចំនួនធំរបស់មនុស្ស ហើយវាបានក្លាយជាវៀងចាំបាច់មិនអាចខ្វះបាន។ ជាមួយគ្នានេះដែរ ការបង្កើតកម្មវិធីកុំព្យូទ័រ គេហទំព័រ ដើម្បីបំពេញនូវតម្រូវការជាក់ស្តែងរបស់មនុស្ស គឺត្រូវបានបង្កើតឡើងដោយការប្រើប្រាស់នូវភាសាកុំព្យូទ័រ(កូដ) និងកម្មវិធីសរសេរខុសៗគ្នាតាមតម្រូវការរបស់អ្នកបង្កើតកម្មវិធី។ ហេតុដូច្នេះហើយបានជា **មជ្ឈមណ្ឌលម៉ាស៊ីនអោយចី** យើងខ្ញុំបានបង្កើតឲ្យមានសៀវភៅ **អ្នកអភិវឌ្ឍន៍ប្រព័ន្ធគ្រប់គ្រងទិន្នន័យ** ដែលជាវគ្គសិក្សាកំពុងពេញនិយមមួយ មានលក្ខណៈស្តង់ដាសម្រាប់បងប្អូនសិក្សានុសិស្សទាំងឡាយចង់ក្លាយជាអ្នកអភិវឌ្ឍន៍ប្រព័ន្ធគ្រប់គ្រងទិន្នន័យ។

គោលដៅនៃការរៀបរៀងសៀវភៅនេះ គឺដើម្បីឲ្យសិស្សនុសិស្ស ក៏ដូចជាអ្នកសិក្សា បានយល់ជ្រាបពី **ការសរសេរប្រព័ន្ធគ្រប់គ្រងទិន្នន័យ** ដោយយើងបានពន្យល់បង្ហាញ ពីធាតុសំខាន់នីមួយៗនៃការអភិវឌ្ឍន៍ប្រព័ន្ធ កម្រិតមធ្យមចំពោះអ្នកដែលបានដឹងខ្លះៗហើយពី HTML, CSS និង JavaScript ចេះប្រើប្រាស់ពីចំណុច ឬក៏ធាតុនីមួយៗ របស់វាដើម្បីយកចំណេះដឹងទាំងអស់នេះទៅបង្កើតជាប្រព័ន្ធគ្រប់គ្រងផ្សេងៗតាមតម្រូវការជាក់ស្តែង។ ក្នុងសៀវភៅនេះយើងផ្តោតសំខាន់ទៅលើការបង្កើតប្រព័ន្ធគ្រប់គ្រងទិន្នន័យការលក់ និងរបាយការណ៍ការលក់ ដើម្បីឲ្យអ្នកសិក្សាឃើញពីសក្តានុពលរបស់ ប្រព័ន្ធគ្រប់គ្រង។ Meteor Framework មិនត្រឹមតែបង្កើតកម្មវិធីប្រព័ន្ធគ្រប់គ្រងបានតែប៉ុណ្ណឹងទេ គេក៏អាចយកវាបង្កើត Website, Window Application, Mobile Application, Game ... និងប្រភេទកម្មវិធីនៅលើ Platform ផ្សេងៗទៀត។

បើទោះជាយ៉ាងណាក៏ដោយសៀវភៅ **អ្នកអភិវឌ្ឍន៍ប្រព័ន្ធគ្រប់គ្រងទិន្នន័យ** គឺសម្រាប់អ្នកដែលមានចំណេះដឹងទៅលើ HTML, CSS, JavaScript កម្រិតមធ្យម។ យើងខ្ញុំសង្ឃឹមថាសិស្សានុសិស្សទាំងអស់គ្នា នឹងបានទទួលចំណេះយ៉ាងពិតប្រាកដក្រោយបញ្ចប់វគ្គសិក្សានេះ។ ជាចុងក្រោយយើងខ្ញុំសុំអភ័យទោសទុកជាមុន ចំពោះប្រការមិនសមរម្យណាមួយដែលកើតមានជាយថាហេតុ នៅក្នុងសៀវភៅមួយក្បាលនេះប៉ុន្តែយើងនឹងទទួលនូវការរិះដើម្បីស្ថាបនាពីសំណាក់អ្នកសិក្សាទាំងឡាយដើម្បីធ្វើការកែតម្រូវឡើងវិញ។ **សូមអរគុណ!!**

រៀបរៀង និងកែសម្រួលដោយ

អ៊ុច បូណា

បោះពុម្ពលើកទី១ ឆ្នាំ២០២១

©រក្សាសិទ្ធិគ្រប់យ៉ាងដោយ **មជ្ឈមណ្ឌលម៉ាស៊ីនអោយចី**



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គួរយល់ដឹង

👉 អ្វីជា Meteor Framework?

វាគឺជា Open Source Framework ដែលផ្តល់ភាពងាយស្រួលក្នុងការបង្កើត និងដាក់ឱ្យប្រើប្រាស់នូវកម្មវិធីច្រើនទម្រង់ដូចជា Web, Mobile និង Desktop ដោយប្រើប្រាស់ភាសា JavaScript។

សព្វថ្ងៃនេះ Meteor Framework មានសក្តានុពលខ្លាំងក្នុងការងារជាច្រើនដែលជាជំនួយដល់មនុស្សជាទូទៅ។

👉 អ្វីជា Vuejs ?

Vuejs គឺជា progressive framework ដែលសម្រាប់រចនា User Interface ។ Vue ត្រូវបានរចនាឡើង ផ្តល់ភាពងាយស្រួលឡើងៗ ពីមួយជំនាន់ទៅមួយជំនាន់។ ប៉ុន្តែវាតម្រូវឱ្យយើងមានសមត្ថភាពទៅលើភាសា HTML, CSS និង JavaScript ក្នុងកម្រិតមធ្យម។

👉 អ្វីជា Quasar Framework?

Quasar គឺជា framework របស់ Vuejs ដែលធ្វើឱ្យយើងជាអ្នកអភិវឌ្ឍន៍គេហទំព័រ មានភាពងាយស្រួលក្នុងការបង្កើត Responsive website ។

យល់ដឹងអំពី Meteor

១. ការដំឡើង

១.១ OSX / Linux

ដោយឡែកនិងធ្វើការដំឡើង Node តាមរយៈ link
<https://nodejs.org/en/download/>

```
curl https://install.meteor.com/ | sh
```

១.២ Windows

ដោយឡែកនិងធ្វើការដំឡើង Node តាមរយៈ link
<https://nodejs.org/en/download/>

```
npm install -g meteor
```

ឬ ធ្វើការដំឡើង Meteor តាមរយៈ Choco ដោយធ្វើការដំឡើង Choco ជាមុន
 សិនដោយធ្វើការ Open PowerShell as administrator រួចធ្វើការ Run script ខាងក្រោម

```
Set-ExecutionPolicy Bypass -Scope Process -  
Force; [System.Net.ServicePointManager]::SecurityProtocol = [System.Net.Service  
PointManager]::SecurityProtocol -bor 3072; iex ((New-  
Object System.Net.WebClient).DownloadString('https://chocolatey.org/install.ps1  
'))
```

បន្ទាប់ពី Run script ចប់សូមធ្វើការ Run script ដើម្បីដំឡើង Meteor ម្តង

- choco install meteor

២. ការបង្កើត Project

```
meteor create myapp --vue
```

៣. ការ Run project

```
cd myapp  
meteor npm install  
meteor
```

៤. VS Code

ជាកម្មវិធីសម្រាប់សរសេរកូដ ដែលផ្តល់ភាពងាយស្រួលដល់ Developer។

ដោយឡែកតាមរយៈ website <https://code.visualstudio.com/>

មេរៀនទី១ ការណែនាំអំពី Vuejs

១.ការប្រកាសអថេរ និង ការបង្ហាញ

១.១ ការប្រកាសអថេរ

គឺជាការតាំងឈ្មោះតំណាងឲ្យតំលៃណាមួយ។ ដើម្បីប្រកាសអថេរយើងត្រូវធ្វើការប្រកាសនៅក្នុង data function។

ឧទាហរណ៍ យើងប្រកាសអថេរឈ្មោះ message រួចយើងផ្តល់តំលៃ "Hello" ឲ្យវា។

```
<script>
  export default {
    data() {
      return {
        message: "Hello",
      };
    },
  };
</script>
```

១.២ ការបង្ហាញ

ដើម្បីបង្ហាញតំលៃរបស់អថេរនៅក្នុង Vue Template បានយើងត្រូវយកអថេរទៅដាក់នៅក្នុងសញ្ញា {{}}

ឧទាហរណ៍ យើងធ្វើការបង្ហាញតំលៃរបស់អថេរដែលមានឈ្មោះ message

```
<template>
  <div>{{ message }}</div>
</template>
```

លទ្ធផល

Hello

១.៣ ការប្រើប្រាស់ V-model

យើងប្រើប្រាស់ v-model សម្រាប់ផ្តល់តំលៃឲ្យអថេរតាមរយៈ form ដោយផ្ទាល់

របៀបប្រើប្រាស់ v-model

```

<template>
  <div>
    <input v-model="message" />
    <h1>{{ message }}</h1>
  </div>
</template>

```

លទ្ធផល

Hello

១.៤ ការប្រើប្រាស់ v-bind

យើងប្រើប្រាស់ v-bind ដើម្បីប្រើប្រាស់តម្លៃរបស់អថេរនៅក្នុង Attribute របស់ HTML។

របៀបប្រើប្រាស់ v-bind

ឧទាហរណ៍ទី១

```

<template>
  <div>
    <input v-bind:value="message" />
    <h1>{{ message }}</h1>
  </div>
</template>
<script>
export default {
  data() {
    return {
      message: "Hello",
    };
  },
};
</script>

```

លទ្ធផល

Hello

ឧទាហរណ៍ទី២ v-model និង v-bind

```
<template>
  <div>
    v-model :<input v-model="message" /><br />
    static value :<input value="This is a static value" /> <br />
    v-bind :<input v-bind:value="message" />
    <h1>{{ message }}</h1>
  </div>
</template>
<script>
export default {
  data() {
    return {
      message: "Hello",
    };
  },
};
</script>
```

លទ្ធផល

v-model : Hello
 static value : This is a static value
 v-bind : Hello

Hello

១.៥ ការប្រើប្រាស់ V-show

យើងប្រើប្រាស់ v-show ដើម្បីបង្ហាញឬលាក់ element របស់ HTML

ឧទាហរណ៍

```
<template>
  <div>
    <h1>
      Message:
      <span v-show="showMessage"> {{ message }} </span>
    </h1>
  </div>
</template>
<script>
  export default {
    data() {
      return {
        message: "Hello",
        showMessage: true,
      };
    },
  };
</script>
```

លទ្ធផល

Message: Hello

លំហាត់

- I. ចូរធ្វើការបង្កើត input field ៤ ដែល field នីមួយៗមាន name, gender, age, phone រួចបង្ហាញដោយប្រើប្រាស់ Default value:

ឈ្មោះ:	ដានីច
ភេទ	ស្រី
អាយុ	២០
លេខទូរស័ព្ទ	017234433

- II. ចូរធ្វើការបង្កើត អញ្ញតដែលមានឈ្មោះថា name, gender, age, phone រួចធ្វើការជំនួសនៅក្នុង input field នីមួយៗតាមរយៈ v-bind:value

=====v-bind=====

ឈ្មោះ:

ភេទ

អាយុ

លេខទូរស័ព្ទ

III. ចូរធ្វើការកែប្រែតម្លៃរបស់អញ្ញតនីមួយៗនៅក្នុង input field តាមរយៈ v-model

=====v-model=====

ឈ្មោះ:

ភេទ

អាយុ

លេខទូរស័ព្ទ

=====v-bind=====

ឈ្មោះ:

ភេទ

អាយុ

លេខទូរស័ព្ទ

Conditionals and Loops

២.លក្ខខណ្ឌ

លក្ខខណ្ឌនៅក្នុងភាសា Programing គឺជាអ្នកកំណត់ Block ក្នុង ណាមួយដំណើរការ ឬ មិនដំណើរការ។

លក្ខខណ្ឌនៅក្នុង Vuejs មាន៣គឺ

- v-if
- v-else if
- v-else

ឧទាហរណ៍

```
<template>
  <div>
    <h5>បញ្ចូលឈ្មោះ: និង អាយុ</h5>
```



```

ឈ្មោះ: <input v-model="name" /><br />
អាយុ <input v-model="age" /><br />
<h5 v-if="age >= 0 && age < 12">អ្នកស្ថិតក្នុងវ័យកុមារ</h5>
<h5 v-else-if="age >= 12 && age < 18">អ្នកស្ថិតក្នុងវ័យជំទង់</h5>
<h5 v-else-if="age >= 18 && age < 24">អ្នកស្ថិតក្នុងវ័យយុវវ័យ</h5>
<h5 v-else>អ្នកស្ថិតក្នុងវ័យចាស់</h5>
</div>
</template>
<script>
export default {
  data() {
    return {
      name: "",
      age: null,
    };
  },
};
</script>

```

លទ្ធផល

បញ្ចូលឈ្មោះ និង អាយុ

ឈ្មោះ:

អាយុ:

អ្នកស្ថិតក្នុងវ័យជំទង់

៣. ទ្វីលដុំ

រឿងលំដុំ គឺជាសកម្មភាពដែលនៅលើ block ក្នុងដែលបង្ហាញធាតុម្តងមួយៗរហូតដល់អស់ធាតុនៃ List ។

ឧទាហរណ៍

```

<template>
  <div>
    <div v-for="(user, index) in users" :key="index">
      {{ user.name }}|{{ user.gender }}|{{ user.age }}
    </div>
  </div>
</template>
<script>
export default {
  data() {
    return {
      users: [

```

```
{
  name: "បុត្រា",
  gender: "ស្រី",
  age: "២០",
},
{
  name: "ប្អូន",
  gender: "ស្រី",
  age: "២១",
},
{
  name: "កែវ",
  gender: "ប្រុស",
  age: "២៣",
},
],
};
},
};
</script>
```

លទ្ធផល

បុត្រា|ស្រី|២០
ប្អូន|ស្រី|២១
កែវ|ប្រុស|២៣

លំហាត់

- I. ចូរសរសេរកូដដើម្បីបោះបង្ហាញតម្លៃរបស់អញ្ញត users រួចធ្វើការច្រោះ(filter)យក user តាមរយៈរោងដោយប្រើប្រាស់ input type radio ដើម្បីធ្វើការច្រោះ(filter) ដូចខាងក្រោម។

☒ ស្រី
☐ ប្រុស
បុត្រា|ស្រី|20
ប្អូន|ស្រី|23

៤. Handling User Input

៤.១ Handling

គឺជាការគ្រប់គ្រង User Input តាមរយៈ directive ដើម្បីភ្ជាប់ event listener សម្រាប់ call ហៅ methods នៅក្នុង Vue។

Directive នៅក្នុង Vue គឺជា Event ដែលយើងប្រើប្រាស់នៅលើ Browser មានដូចជា click, enter ។ល។

Attribute គឺជា parameter របស់ Directive។ យើងសំគាល់វាបានដោយមើលទៅលើសញ្ញា colon (:) ខាងឆ្វេង colon គឺជា directive ចំណែកខាងស្តាំ colon គឺជា parameter ។

ឧទាហរណ៍

```
<template>
  <div>
    <h4>{{ username }}</h4>
    <button v-on:click="reverseName">Reverse</button>
    <button v-on:click="convertToUpperCase">ToUppercase</button>
    <button v-on:click="convertToLowerCase">ToLowerCase</button>
  </div>
</template>
<script>
export default {
  data() {
    return {
      username: "Coravid",
    };
  },
  methods: {
    reverseName() {
      this.username = this.username.split("").reverse().join("");
    },
    convertToUpperCase() {
      this.username = this.username.toUpperCase();
    },
    convertToLowerCase() {
      this.username = this.username.toLowerCase();
    },
  },
};
</script>
```

លទ្ធផល

Coravid

Reverse
ToUppercase
ToLowerCase

៤.២ User Input

ឧទាហរណ៍

```

<template>
  <div>
    enter : <input v-model="username" v-
on:keydown.enter="handleEnter" /> <br />
    esc: <input v-model="username" v-
on:keydown.esc="handleEsc" />
  </div>
</template>
<script>
export default {
  data() {
    return {
      username: "Marina",
    };
  },
  methods: {
    handleEnter() {
      alert(this.username);
    },
    handleEsc() {
      this.username = null;
    },
  },
};
</script>

```

លទ្ធផល

enter :

esc:

Methods

ដើម្បីសរសេរ Functions បានយើងត្រូវសរសេរនៅក្នុង methods object របស់ Vue។

៥. ទម្រង់កាត់របស់ v-bind និង v-on

ទម្រង់កាត់របស់ v-bind គឺសញ្ញា (:)

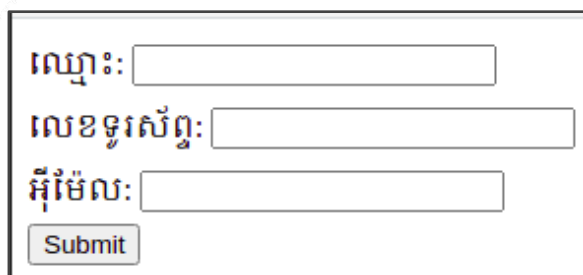
ឧទាហរណ៍ v-bind: title = :title

ទម្រង់កាត់របស់ v-on គឺសញ្ញា (@)

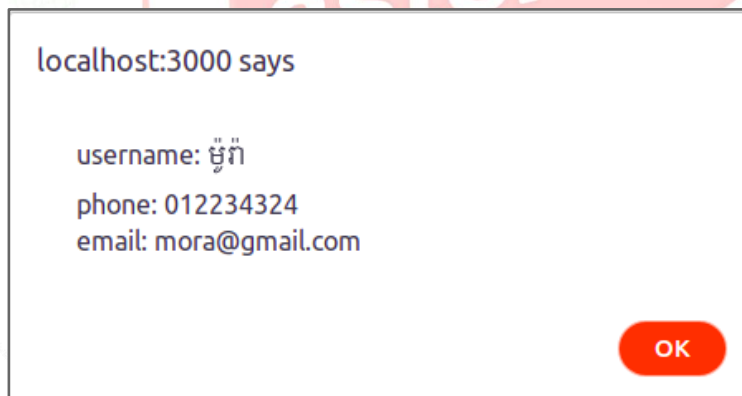
ឧទាហរណ៍ v-on: click = @click

លំហាត់

I. ចូរបង្កើត Form មួយដែលមាន ៣ Field និង ប៊ូតុង submit មួយដូចខាងក្រោម



ហើយនៅពេលដែលយើងចុចប៊ូតុង submit យើងនឹងទទួលបាន alert មួយនៅលើ browser ដូចខាងក្រោម



៥.១ Computed Properties and Watchers

ក. Computed

Computed គឺជា Object មួយនៅក្នុង Vue Instant។ វាគឺជាអ្នកសង្កេតមើលនិងធ្វើការទៅលើ data ពេលណាដែល data ត្រូវបាន ប្រែប្រួល។ ឧទាហរណ៍

```
<script>
export default {
  data() {
    return {
      firstname: "jame",
      lastname: "maron",
    }
  }
}
```

```
};
},
computed: {
  fullname() {
    return this.firstname + " " + this.lastname;
  },
},
};
</script>
```

ខ. Watcher

Watcher គឺជាអ្នកសង្កេតមើលនិងធ្វើការទៅលើ data ពេលណាដែល data ត្រូវបានប្រែប្រួល។ ឧទាហរណ៍

```
<script>
export default {
  data() {
    return {
      firstname: "jame",
      lastname: "maron",
      fullname: "jame maron",
    };
  },
  watch: {
    firstname: function (val) {
      this.fullname = val + " " + this.lastname;
    },
    lastname: function (val) {
      this.fullname = this.firstname + " " + val;
    },
  },
};
</script>
```

តើ computed properties និង watcher មានភាពខុសគ្នាយ៉ាងដូចម្តេច?

គ. Instance Lifecycle Hooks

- Before Create
- created
- before Mount
- mounted
- before Update
- updated
- before Destroy
- destroyed

ឧទាហរណ៍

```

<template>
  <div>
    <user v-if="tab"></user>
    <item v-else></item>
    <button @click="handleNext()">Next</button>
  </div>
</template>
<script>
import Item from "../components/Item.vue";
import User from "../components/User.vue";

export default {
  components: { User, Item },
  mounted() {
    console.log("mounted:");
  },
  beforeMount() {
    console.log("beforeMount:");
  },
  created() {
    console.log("created:");
  },
  beforeCreate() {
    console.log("beforeCreate:");
  },
  updated() {
    console.log("updated:");
  },
  beforeUpdate() {
    console.log("beforeUpdate");
  },
  destroyed() {
    console.log("destroyed:");
  },
  beforeDestroy() {
    console.log("beforeDestroy:");
  },
  data() {
    return {
      tab: false,
    };
  },
  methods: {
    handleNext() {
      this.tab = !this.tab;
    },
  },
};
</script>

```

លំហាត់

- I. ចូរសរសេរកម្មវិធីមួយដែលអាចឲ្យគេបញ្ចូលតម្លៃ name, price, quantity និងប្រើប្រាស់ computed ដើម្បីគណនារក amount រួចធ្វើការ Push តម្លៃទាំងនោះ ចូល Array មួយ មានឈ្មោះថា items រួចធ្វើការបង្ហាញ មកវិញ។

name:

price:

quantity:

amount: 6.25

coca | 0.3 | 4

abc | 1.25 | 5

៥.២ Class and Style Bindings

ក. Class binding

Class binding គឺជាការ bind class ទៅតាមតម្លៃដែលគេផ្តល់ឲ្យ។ តម្លៃរបស់ class binding មានតែ២ប៉ុណ្ណោះ គឺ true និង false។

ឧទាហរណ៍ទី១

```

<template>
  <div>
    <h4 class="static-class">This is a static class</h4>
    <h4 :class="{ 'binding-class': active }">
      This is a binding class( dynamic class)
    </h4>
    <button @click="toggle">Toggle dynamic class</button>
  </div>
</template>
<script>
export default {
  data() {
    return {
      active: false,
    };
  },
  methods: {
    toggle() {
      this.active = !this.active;
    },
  },
};
</script>
<style>
.static-class {
  text-align: center;

```

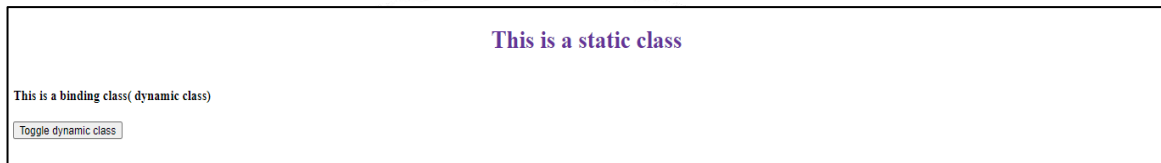


```

    color: rebeccapurple;
    font-size: 30px;
  }
  .binding-class {
    text-align: center;
    color: blue;
  }
</style>

```

លទ្ធផល



ឧទាហរណ៍ទី២

```

<template>
  <div>
    <h4 class="static-class">This is a static class</h4>
    <h4 :class="objectClass">This is a binding class( dynamic clas
s)</h4>
    <button @click="toggle">Toggle dynamic class</button>
  </div>
</template>
<script>
export default {
  data() {
    return {
      objectClass: {
        font: true,
        "text-danger": false,
      },
    };
  },
  methods: {
    toggle() {
      this.objectClass["text-danger"] = true;
    },
  },
};
</script>
<style>
.static-class {
  text-align: center;
  color: rebeccapurple;
  font-size: 30px;
}
.text-danger {
  color: red;
}

```

```

    }
    .font {
      font-size: 30px;
    }
  </style>

```

លទ្ធផល

This is a static class

This is a binding class(dynamic class)

Toggle dynamic class

១. Style binding

ទម្រង់សម្រាប់ Object គឺ v-bind:style វាងាយស្រួលយល់ជាងការដាក់ CSS ដោយផ្ទាល់ ជាទម្រង់ JavaScript Object. អ្នកអាចប្រើប្រាស់ទម្រង់ camelCase ឬ kebab-case (ដោយប្រើ single quotes with kebab-case) សម្រាប់ឈ្មោះរបស់ CSS Properties:

ឧទាហរណ៍

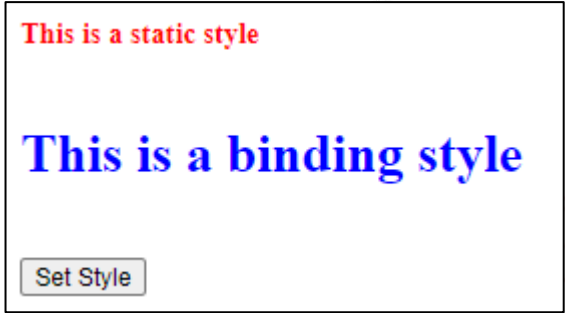
```

<template>
  <div>
    <h4 style="color: red">This is a static style</h4>
    <h4 :style="objectStyle">This is a binding style</h4>
    <button @click="handleSetStyle">Set Style</button>
  </div>
</template>
<script>
export default {
  data() {
    return {
      objectStyle: {
        fontSize: "30px",
        color: "blue",
      },
    };
  },
  methods: {
    handleSetStyle() {
      if (this.objectStyle.fontSize == "70px") {
        this.objectStyle = {
          fontSize: "30px",
          color: "blue",
        };
      }
    }
  }
};

```

```
    } else {
      this.objectStyle.fontSize = "70px";
      this.objectStyle.color = "pink";
    }
  },
},
};
</script>
```

លទ្ធផល



លំហាត់

- I. ចូរបង្កើត Form មួយដែលមាន Field ដូចខាងក្រោមនៅពេលដែលចុចប៊ូតុង Sign up យើងត្រូវយកទិន្នន័យពី Form ទៅបញ្ចូលក្នុង Array មួយដែលមានឈ្មោះថា users រួចធ្វើការបង្ហាញទិន្នន័យរបស់ users ដោយប្រើប្រាស់ Binding Class ក្នុង Table។

Sign up

Username :

password :

confirm :

Gender: Male ☒ Female ☐

Habits: hiking ☒ football ☒ reading ☐ eating ☐ hangout ☐

Sign up

Username	Gender	Habits
Biona	male	["hiking", "football"]

៥.៣ Components

អ្វីទៅជា Components ?

Components ជាការបង្កើត element របស់ html tag ឬអាចហៅម្យ៉ាងទៀតបានថាជា Tools សម្រាប់ប្រើប្រាស់ច្រើនដង។ វាមានលក្ខណៈដូច Function ដែរដែលអនុញ្ញាតឱ្យយើងហៅវាប្រើបានច្រើនកន្លែង។

ក. Components គ្មាន parameter

➤ របៀបបង្កើត Components

យើងត្រូវបង្កើត File រួចដាក់ឈ្មោះឱ្យ Component នោះ

ឧទាហរណ៍

->project/imports/ui/components/MyButton.vue

```
<template>
  <div>
    <button class="button button__text">Enter Button</
button>
  </div>
</template>

<style>
.button {
  position: relative;
  padding: 0;
  width: 200px;
  height: 200px;
  border: 4px solid #888888;
  outline: none;
  background-color: #f4f5f6;
  border-radius: 40px;
  box-shadow: -6px -20px 35px #ffffff, -6px -
10px 15px #ffffff,
  -
20px 0px 30px #ffffff, 6px 20px 25px rgba(0, 0, 0, 0.2);
  transition: 0.13s ease-in-out;
  cursor: pointer;
}
.button:active {
  box-shadow: none;
}
.button:active .button__content {
  box-shadow: none;
}
.button:active .button__content .button__text,
```

```

.button:active .button__content .button__icon {
  transform: translate3d(0px, 0px, 0px);
}
.button__content {
  position: relative;
  display: grid;
  padding: 20px;
  width: 100%;
  height: 100%;
  grid-template-columns: 1fr 1fr 1fr 1fr;
  grid-template-rows: 1fr 1fr;
  box-shadow: inset 0px -8px 0px #dddddd, 0px -
8px 0px #f4f5f6;
  border-radius: 40px;
  transition: 0.13s ease-in-out;
  z-index: 1;
}
.button__icon {
  position: relative;
  display: flex;
  transform: translate3d(0px, -4px, 0px);
  grid-column: 4;
  align-self: start;
  justify-self: end;
  width: 32px;
  height: 32px;
  transition: 0.13s ease-in-out;
}
.button__icon svg {
  width: 32px;
  height: 32px;
  fill: #aaaaaa;
}
.button__text {
  position: relative;
  transform: translate3d(0px, -4px, 0px);
  margin: 0;
  align-self: end;
  grid-column: 1/5;
  grid-row: 2;
  text-align: center;
  font-size: 32px;
  background-color: #888888;
  color: transparent;
  text-shadow: 2px 2px 3px rgba(255, 255, 255, 0.5);
  -webkit-background-clip: text;
  -moz-background-clip: text;
  background-clip: text;
  transition: 0.13s ease-in-out;
}

```

```
.credits {
  margin-top: 24px;
}
.credits__reference {
  display: inline-block;
  border-bottom: 1px solid transparent;
  color: #0099ff;
  text-decoration: none;
  transition: ease-in 0.13s;
}
.credits__reference:hover {
  border-bottom-color: #0099ff;
}
</style>
```

➤ របៀបហៅ component មកប្រើប្រាស់

ទី១ យើងត្រូវ import file component ដែលយើងបានបង្កើត

ឧទាហរណ៍

```
import MyButton from "../components/MyButton.vue";
```

ទី២ យើងត្រូវយកឈ្មោះ File ដែលយើងបាន import មកយកទៅដាក់ក្នុង component object របស់ Vue។

ឧទាហរណ៍

```
components: { MyButton },
```

ទី៣ យើងត្រូវអាចហៅវាប្រើបាននៅក្នុង Vue template ដូចជា tag ធម្មតាដែរ

ឧទាហរណ៍

```
<template>
  <div>
    <my-button /> <br />
    <my-button /> <br />
  </div>
</template>
```

១.Components មាន parameter

របៀបបោះពុម្ព parameter

```
<my-button title="Submit" /> <br />
```

របៀបចាប់តម្លៃ parameter

```

<script>
export default {
  props: {
    title: {
      type: String,
      default: "Button",
    },
  },
};
</script>

```

គ.ការផ្តល់តម្លៃត្រឡប់តាមរយៈ Event

Syntax

```

this.$emit("event", value)
<my-button @event="eventName" />

```

ឧទាហរណ៍

ក្នុង App.vue

```

<template>
  <div>
    <my-button @submit="onSubmit" />
  </div>
</template>
<script>
import MyButton from "./components/MyButton.vue";
export default {
  components: { MyButton },
  methods: {
    onSubmit(username) {
      alert("Hello!" + username);
    },
  },
};
</script>

```

ក្នុង component/MyButton.vue

```

<template>
  <div>
    <button @click="handleSubmit">My Button</button>
  </div>

```

```

</template>
<script>
export default {
  methods: {
    handleSubmit() {
      let username = "Biona";
      this.$emit("submit", username);
    },
  },
};
</script>

```

លទ្ធផល



ពេលចុចត្រង់ Button នោះវានឹង alert ដូចខាងក្រោម



លំហាត់

- I. ចូរបង្កើត Component Signup និង Component Table រួចធ្វើការហៅប្រើនៅក្នុង Root(App.vue) ដូចខាងក្រោម



Username :

password :

confirm :

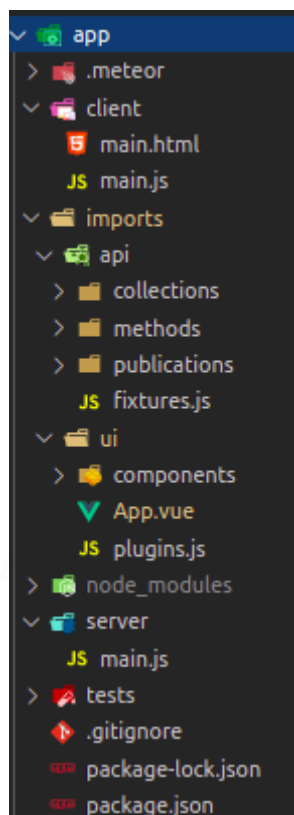
Gender: Male ☒ Female ☐

Habits: hiking ☒ football ☒ reading ☐ eating ☐ hangout ☐

Username	Gender	Habits
Biona	male	["football", "hiking"]

មេរៀនទី២ ការណែនាំអំពី Meteorjs, Quasar, MongoDB

Meteor Structure



- .meteor/ # source meteor framework
- client/ # All code in the client directory is run only in the client-side or web browser.
 - main.js # main JavaScript file
 - main.html # main html file
- import/ #
 - api/ #
 - collections # កន្លែងបង្កើត collections នៅលើ server
 - methods # កន្លែងសរសេរកូដ method នៅលើ server
 - publications # all related publications
 - ui/ #
 - components/ # All reusable components in the application
 - App.vue # Root app of Vue
 - plugins.js # configure plugins
- node_modules/ # node package
- server/
 - main.js # server entry point, imports all server code
- package.json # list package name

១. Quasar

១.១ ការដំឡើង

ដើម្បីប្រើប្រាស់ Quasar បានយើងត្រូវដំឡើងវានៅក្នុង Project របស់យើង។
របៀបដំឡើងមាន២របៀបគឺ៖ ទី១ ការដំឡើងតាមរយៈ CDN(Content Network Delivery)

ទី២ ការដំឡើងតាមរយៈ NPM(Node Package Manager)

```
npm install --save quasar@1.15.21
npm i @quasar/extras
```

១.២ Configuration Quasar

របៀបក្នុងការ configuration យើងត្រូវចូលទៅសរសេរនៅក្នុង File

->imports/ui/plugins.js

```
import Quasar from "quasar";
Vue.use(Quasar);
import "../public/material-icons.css";
import "quasar/dist/quasar.min.css";
```

យើងត្រូវបង្កើត folder public នៅក្នុង root project រួចចូលទៅ copy file នៅក្នុង folder

->node_modules/@quasar/extras/material-icons ទាំងអស់ដាក់ក្នុង folder
->public

Vue Router

Vue Router គឺជា Feature មួយរបស់ Vue ប្រើសម្រាប់ Navigate ពី Page មួយទៅ Page មួយទៀត។

១.៣ របៀបដំឡើង Vue Router

```
npm install vue-router
```

របៀប Config

បង្កើត file

->imports/ui/pages/Home.vue

```
<template>
  <div>
```

```

    <h1>Home</h1>
  </div>
</template>

```

->imports\ui\pages\Customer.vue

```

<template>
  <div>
    <h1>Customer</h1>
  </div>
</template>

```

ត្រូវបង្កើត folder និង file ->imports/ui/routes/index.js រួចធ្វើការសរសេរ ដូចខាងក្រោម

```

import Vue from "vue";
import VueRouter from "vue-router";
Vue.use(VueRouter);
let routes = [
  {
    path: "/",
    component: () => import("../pages/Home.vue"),
    name: "home",
  },
  {
    path: "/user",
    component: () => import("../pages/User.vue"),
    name: "user",
  },
  {
    path: "/customer",
    component: () => import("../pages/Customer.vue"),
    name: "customer",
  },
];
const router = new VueRouter({
  routes,
  mode: "history",
});
export default router;

```

រួចធ្វើការ Import file router ចូលទៅក្នុង file ->client/main.js

```

import Vue from "vue";
import "../imports/ui/plugins";

import App from "../imports/ui/App.vue";
import router from "../imports/ui/routes";
Meteor.startup(() => {
  new Vue({

```

```
el: "#app",
router,
...App,
});
});
```

១.៤ អនុវត្ត

ក. Layout

-> imports/ui/App.vue

```
<template>
  <div>
    <q-layout view="hHr LpR lfr">
      <q-header class="bg-primary text-white" elevated>
        <q-toolbar>
          <q-
            btn flat @click="drawer = !drawer" round dense icon="menu" />
          <q-toolbar-title>Master IT Shop</q-toolbar-title>
        </q-toolbar>
      </q-header>

      <q-drawer v-model="drawer" content-class="bg-grey-3" show-if-
        above
      >

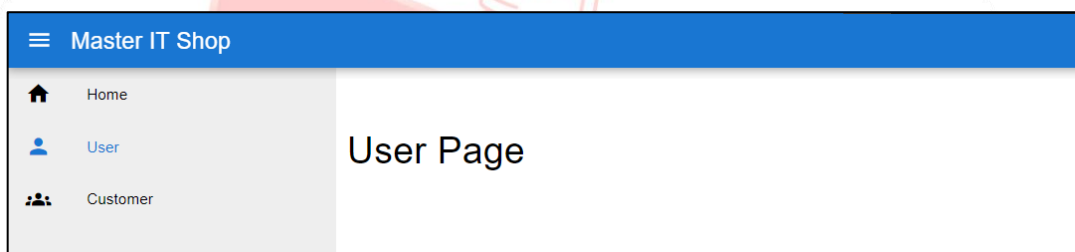
        <q-scroll-area class="fit">
          <q-list>
            <q-item clickable v-ripple exact to="/">
              <q-item-section avatar>
                <q-icon name="home" />
              </q-item-section>
              <q-item-section> Home </q-item-section>
            </q-item>
            <q-item clickable v-ripple to="/user">
              <q-item-section avatar>
                <q-icon name="person" />
              </q-item-section>
              <q-item-section> User </q-item-section>
            </q-item>
            <q-item clickable v-ripple to="/customer">
              <q-item-section avatar>
                <q-icon name="groups" />
              </q-item-section>
              <q-item-section> Customer </q-item-section>
            </q-item>
          </q-list>
        </q-scroll-area>
      </q-drawer>
      <q-page-container>
        <q-page>
```

```

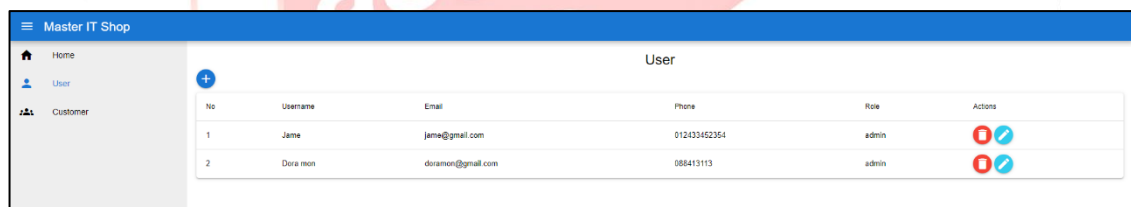
        <router-view />
      </q-page>
    </q-page-container>
  </q-layout>
</div>
</template>
<script>
export default {
  data() {
    return {
      drawer: false,
    };
  },
};
</script>

```

លទ្ធផល



១. User



Username
Dora

Email
dora@gmail.com

Phone
02354353

Role
Admin

Password

Confirm

SUBMIT

ការបង្កើត Page

-> imports\ui\pages\User.vue

```

<template>
  <div>
    <q-dialog
      v-model="dialog"
      persistent
      :maximized="true"
      transition-show="slide-up"
      transition-hide="slide-down"
    >
      <q-card>
        <q-bar>
          <q-space />
          <q-btn
            dense
            flat
            icon="close"
            color="white"
            class="bg-red shadow-4"
            rounded
            v-close-popup
          >
            <q-tooltip content-class="bg-white text-
primary ">Close</q-tooltip>
          </q-btn>
        </q-bar>
        <q-card-section>
          <user-form :updateDoc="updateDoc" @close="close" />
        </q-card-section>
      </q-card>
    </q-dialog>
    <q-btn
      icon="add"
      class="q-mb-sm"
      dense
      color="primary"
      rounded
      @click="handleAdd"
    />
    <user-
table :users="users" @delete="handleDelete" @edit="handleEdit" />
  </div>
</template>

<script>
import UserForm from "../components/UserForm.vue";
import UserTable from "../components/UserTable.vue";
export default {
  components: { UserTable, UserForm },
  data() {
    return {
      dialog: false,

```

```

        updateDoc: null,

        users: [],
    };
},
mounted() {
    this.getData();
},
methods: {
    handleAdd() {
        this.dialog = true;
    },
    close() {
        this.dialog = false;
        this.updateDoc = null;
        this.getData();
    },
    getData() {
        this.users = [
            {
                _id: "01",
                username: "Jame",
                email: "jame@gmail.com",
                phone: "012433452354",
                role: "admin",
            },
            {
                _id: "02",
                username: "Dora mon",
                email: "doramon@gmail.com",
                phone: "088413113",
                role: "admin",
            },
        ];
    },
    handleDelete(id) {
        let index = this.users.findIndex((doc) => {
            return doc._id == id;
        });
        this.users.splice(index, 1);
    },
    handleEdit(doc) {
        this.updateDoc = Object.assign({}, doc);
        this.dialog = true;
        console.log('doc', doc);
    },
},
};
</script>

```

ការបង្កើត Component Form

->imports\ui\components\UserForm.vue

```
<template>
  <div>
    <div class="row justify-around q-ma-sm">
      <div class="col-md-5 col-xs-12 col-sm-4 q-ma-md">
        <q-input outlined v-
model="form.username" label=" Username" />
      </div>
      <div class="col-md-5 col-xs-12 col-sm-4 q-ma-md">
        <q-input outlined v-model="form.email" label=" Email" />
      </div>
      <div class="col-md-5 col-xs-12 col-sm-4 q-ma-md">
        <q-input outlined v-model="form.phone" label=" Phone" />
      </div>
      <div class="col-md-5 col-xs-12 col-sm-4 q-ma-md">
        <q-select
          outlined
          v-model="form.role"
          :options="roles"
          label="Role"
          map-options
          emit-value
          option-value="value"
        />
      </div>
      <div class="col-10 q-ma-md" v-if="updateDoc">
        <q-checkbox v-model="reset" label="Reset Password" />
      </div>
    </div>
    <div class="row justify-around q-ma-sm" v-
if="!updateDoc || reset">
      <div class="col-md-5 col-xs-12 col-sm-4 q-ma-md">
        <q-input
          outlined
          v-model="form.password"
          type="password"
          label="Password"
        />
      </div>
      <div class="col-md-5 col-xs-12 col-sm-4 q-ma-md">
        <q-input outlined v-
model="confirm" type="password" label="Confirm" />
      </div>
    </div>
    <div class="q-ma-md text-center">
      <q-btn
        :color="updateDoc ? `info` : `primary`"
        :label="updateDoc ? `Update` : `Submit`"
        @click="handleSubmit()"
      />
    </div>
  </div>
</template>
```



```

    </div>
  </div>
</template>

<script>
export default {
  props: {
    updateDoc: {
      default: null,
      type: Object,
    },
  },
  data() {
    return {
      form: {
        username: "Dora",
        email: "dora@gmail.com",
        phone: "02354353",
        role: "admin",
        password: "dora",
      },
      reset: false,
      confirm: null,
      roles: [
        {
          label: "Admin",
          value: "admin",
        },
        {
          label: "Sale",
          value: "sale",
        },
      ],
    };
  },
  mounted() {
    if (this.updateDoc) {
      this.form = this.updateDoc;
    }
  },
  methods: {
    handleSubmit() {
      if (this.form.password == this.confirm && this.form.password !=
null) {
        if (this.reset) {
          // code insert
          this.form.reset = this.reset;
        }
      }
      if (this.updateDoc) {

```

```
// code update
}
this.$emit("close");
console.log("form:", this.form);
},
},
};
</script>
```

ការបង្កើត Component Table

->imports\ui\components\CustomerTable.vue

```
<template>
  <div>
    <q-markup-table>
      <thead>
        <tr>
          <th class="text-left">No</th>
          <th class="text-left">Username</th>
          <th class="text-left">Email</th>
          <th class="text-left">Phone</th>
          <th class="text-left">Role</th>
          <th class="text-left">Actions</th>
        </tr>
      </thead>
      <tbody>
        <tr v-for="(user, index) in users" :key="user._id">
          <td class="text-left">{{ index + 1 }}</td>
          <td class="text-left">{{ user.username }}</td>
          <td class="text-left">{{ user.email }}</td>
          <td class="text-left">{{ user.phone }}</td>
          <td class="text-left">{{ user.role }}</td>
          <td class="text-left">
            <q-btn
              color="red"
              dense
              icon="delete"
              rounded
              @click="handleDelete(user._id)"
            />
            <q-btn
              color="info"
              dense
              icon="edit"
              rounded
              @click="habndleEdit(user)"
            />
          </td>
        </tr>
      </tbody>
    </q-markup-table>
```

```

</div>
</template>

<script>
export default {
  props: {
    users: {
      type: Array,
      default: null,
      required: true,
    },
  },
  methods: {
    handleDelete(id) {
      this.$emit("delete", id);
    },
    handleEdit(doc) {
      this.$emit("edit", doc);
    },
  },
};
</script>

```

៥ Customer

The screenshot shows a web application interface for a 'Master IT Shop'. On the left is a sidebar with navigation links: 'home', 'User', and 'Customer'. The main area displays a 'Customer' table with the following data:

No	Name	Gender	Dob	Phone	Address	Status	Actions
1	Jane	male	02-04-1997	010202022	battambang	active	 
2	Bisma	female	27-05-1999	017222277	battambang	active	 

Below the table, a 'Customer Form' modal is open. It contains the following fields:

- Name:
- Date of Birth: (with a calendar icon)
- Address:
- Gender: ☒ Male ☐ Female
- Phone:
- Status: ☒ Active ☐ Inactive

A 'SUBMIT' button is located at the bottom left of the form.

ការបង្កើត Page

-> imports\ui\pages\Customer.vue

ដំឡើង Moment js

```
npm i moment
```

```

<template>
  <div>
    <q-dialog

```

```

        v-model="dialog"
        persistent
        :maximized="true"
        transition-show="slide-up"
        transition-hide="slide-down"
      >
        <q-card>
          <q-bar>
            <q-space />
            <q-btn
              dense
              flat
              icon="close"
              color="white"
              class="bg-red shadow-4"
              rounded
              v-close-popup
            >
              <q-tooltip content-class="bg-white text-
primary ">Close</q-tooltip>
            </q-btn>
          </q-bar>
          <customer-form :updateDoc="updateDoc" @close="close" />
        </q-card>
      </q-dialog>
      <q-btn
        icon="add"
        class="q-mb-sm"
        dense
        color="primary"
        rounded
        @click="handleAdd"
      />
      <customer-table
        @edit="handleEdit"
        @delete="handleDelete"
        :customers="customers"
      />
    </div>
  </template>

  <script>
import moment from "moment";
import CustomerForm from "../components/CustomerForm.vue";
import CustomerTable from "../components/CustomerTable.vue";
export default {
  components: {
    CustomerForm,
    CustomerTable,
  },
  data() {

```

```

    return {
      dialog: false,
      updateDoc: null,
      customers: [],
    };
  },
  mounted() {
    this.getData();
  },
  methods: {
    close(doc) {
      if (doc?._id) {
      } else {
        doc._id = new Date();
        this.customers.push(doc);
      }
      console.log("close:", doc);
      this.updateDoc = null;
      this.dialog = false;
    },
    handleDelete(id) {
      console.log("delete on :", id);
      let index = this.customers.findIndex((doc) => {
        return doc._id == id;
      });
      this.$delete(this.customers, index);
    },
    handleEdit(doc) {
      console.log("customer:", doc);
      doc.dob = moment(doc.dob).format("YYYY-MM-DD");
      this.updateDoc = Object.assign({}, doc);
      this.dialog = true;
    },
    handleAdd() {
      this.dialog = true;
    },
    getData() {
      this.customers.push(
        {
          _id: "01",
          name: "Jame",
          gender: "male",
          dob: moment("02-04-1997", "DD-MM-YYYY").toDate(),
          phone: "010202022",
          address: "battambang",
          status: "active",
        },
        {
          _id: "02",
          name: "Biona",

```

```
gender: "female",
dob: moment("27-05-1999", "DD-MM-YYYY").toDate(),
phone: "017222277",
address: "battambang",
status: "active",
  }
);
},
},
};
</script>
```

ការបង្កើត Component Form

->imports\ui\components\CustomerForm.vue

```
<template>
  <div class="q-pr-xl q-pl-xl q-pt-md q-mr-xl q-ml-xl">
    <q-card class="shadow-6">
      <q-card-section class="text-h5 text-center">
        Customer Form
      </q-card-section>
      <q-card-section>
        <div class="fit row justify-evenly">
          <div class="col-xs-12 col-sm-4 q-ma-sm">
            <q-input square outlined v-
model="form.name" label="Name" />
          </div>
          <div class="col-xs-12 col-sm-4 q-ma-sm">
            <fieldset>
              <label class="text-grey text-subtitle1">Gender</label>
              <q-radio v-
model="form.gender" val="male" label="Male" />
              <q-radio v-
model="form.gender" val="female" label="Female" />
            </fieldset>
          </div>
          <div class="col-xs-12 col-sm-4 q-ma-sm">
            <q-input
              type="date"
              square
              outlined
              v-model="form.dob"
              label="Date Of Birth"
              hint="MM/DD/YYYY"
            />
          </div>
          <div class="col-xs-12 col-sm-4 q-ma-sm">
            <q-input square outlined v-
model="form.phone" label="Phone" />
          </div>
          <div class="col-xs-12 col-sm-4 q-ma-sm">
```

```

        <q-input square outlined v-
model="form.address" label="Address" />
      </div>
      <div class="col-xs-12 col-sm-4 q-ma-sm">
        <fieldset>
          <label class="text-grey text-subtitle1">Status</label>
          <q-radio v-
model="form.status" val="active" label="Active" />
          <q-radio v-
model="form.status" val="inactive" label="Inactive" />
        </fieldset>
      </div>
      <div class="col-xs-12 col-sm-4 q-ma-sm">
        <q-btn
          :color="updateDoc ? `info` : `primary`"
          :label="updateDoc ? `Update` : `Submit`"
          @click="handleSubmit"
        />
      </div>
    <div class="col-xs-12 col-sm-4 q-ma-sm"></div>
  </div>
</q-card-section>
</q-card>
</div>
</template>

<script>
import moment from "moment";
export default {
  props: {
    updateDoc: {
      type: Object,
      default: null,
    },
  },
  data() {
    return {
      form: {
        name: null,
        gender: "male",
        dob: moment(new Date()).subtract(18, "years").format("YYYY-
MM-DD"),
        phone: null,
        address: null,
        status: "active",
      },
    };
  },
  created() {
    if (this.updateDoc) {

```

```

        this.form = this.updateDoc;
      }
    },
    methods: {
      handleSubmit() {
        this.form.dob = moment(this.form.dob, "YYYY-MM-DD").toDate();
        console.log("form:", this.form);
        this.$emit("close", this.form);
      },
    },
  },
};
</script>

```

ការបង្កើត Component Table

->imports\ui\components\CustomerTable.vue

```

<template>
  <div>
    <q-markup-table>
      <thead>
        <tr>
          <th class="text-left">No</th>
          <th class="text-left">Name</th>
          <th class="text-left">Gender</th>
          <th class="text-left">Dob</th>
          <th class="text-left">Phone</th>
          <th class="text-left">Address</th>
          <th class="text-left">Status</th>
          <th class="text-center">Actions</th>
        </tr>
      </thead>
      <tbody>
        <tr v-for="(customer, index) in customers" :key="index">
          <td class="text-left">{{ index + 1 }}</td>
          <td class="text-left">{{ customer.name }}</td>
          <td class="text-left">{{ customer.gender }}</td>
          <td class="text-left">{{ customer.dob | toDate }}</td>
          <td class="text-left">{{ customer.phone }}</td>
          <td class="text-left">{{ customer.address }}</td>
          <td class="text-left">{{ customer.status }}</td>
          <td class="text-center">
            <q-btn
              icon="delete"
              dense
              rounded
              color="red"
              flat
              @click="handleDelete(customer._id)"
            />
            <q-btn
              icon="edit"

```



```

        dense
        rounded
        color="info"
        flat
        @click="handleEdit(customer)"
      />
    </td>
  </tr>
</tbody>
</q-markup-table>
</div>
</template>

<script>
import moment from "moment";
export default {
  props: {
    customers: {
      type: Array,
      default: null,
    },
  },
  filters: {
    toDate(date) {
      return moment(date).format("DD-MM-YYYY");
    },
  },
  methods: {
    handleDelete(id) {
      this.$emit("delete", id);
    },
    handleEdit(doc) {
      this.$emit("edit", doc);
    },
  },
};
</script>

```

៣. Category

Master IT Shop					
Home User Customer Category + Category					
No	Name	Date	Description	Actions	
1	Drink	25-06-2021	soft-drink		

ការបន្ថែម Routes

->imports/ui/routes/index.js

```
import Vue from "vue";
import VueRouter from "vue-router";
Vue.use(VueRouter);
let routes = [
  {
    path: "/",
    component: () => import("../pages/Home.vue"),
    name: "Home",
  },
  {
    path: "/user",
    component: () => import("../pages/User.vue"),
    name: "User",
  },
  {
    path: "/customer",
    component: () => import("../pages/Customer.vue"),
    name: "Customer",
  },
  {
    path: "/category",
    component: () => import("../pages/Category.vue"),
    name: "Category",
  },
];
const router = new VueRouter({
  routes,
  mode: "history",
});
export default router;
```

ការបន្ថែម Menus

->imports/ui/App.vue

```
<template>
  <div>
    <q-layout view="hHr LpR lfr">
```

```

    <q-header class="bg-primary text-white" elevated>
      <q-toolbar>
        <q-
btn flat @click="drawer = !drawer" round dense icon="menu" />
        <q-toolbar-title>Master IT Shop</q-toolbar-title>
      </q-toolbar>
    </q-header>

    <q-drawer v-model="drawer" content-class="bg-grey-3" show-if-
above>
      <q-scroll-area class="fit">
        <q-list>
          <q-item clickable v-ripple exact to="/">
            <q-item-section avatar>
              <q-icon name="home" />
            </q-item-section>
            <q-item-section> Home </q-item-section>
          </q-item>
          <q-item clickable v-ripple to="/user">
            <q-item-section avatar>
              <q-icon name="person" />
            </q-item-section>
            <q-item-section> User </q-item-section>
          </q-item>
          <q-item clickable v-ripple to="/customer">
            <q-item-section avatar>
              <q-icon name="groups" />
            </q-item-section>
            <q-item-section> Customer </q-item-section>
          </q-item>
          <q-item clickable v-ripple to="/category">
            <q-item-section avatar>
              <q-icon name="backup_table" />
            </q-item-section>
            <q-item-section> Category </q-item-section>
          </q-item>
        </q-list>
      </q-scroll-area>
    </q-drawer>

    <q-page-container>
      <q-page class="q-mt-md q-ml-md q-mr-md">
        <div class="text-center text-h5">{{ $route.name }}</div>
        <router-view />
      </q-page>
    </q-page-container>
  </q-layout>
</div>
</template>

```

ការបង្កើត Page Category

->imports/ui/pages/Category.vue

```

<template>
  <div>
    <q-dialog
      v-model="dialog"
      persistent
      :maximized="true"
      transition-show="slide-up"
      transition-hide="slide-down"
    >
      <q-card>
        <q-bar>
          <q-space />
          <q-btn
            dense
            flat
            icon="close"
            color="white"
            class="bg-red shadow-4"
            rounded
            v-close-popup
          >
            <q-tooltip content-class="bg-white text-
primary ">Close</q-tooltip>
          </q-btn>
        </q-bar>
        <category-form :updateDoc="updateDoc" @close="close" />
      </q-card>
    </q-dialog>
    <q-btn
      icon="add"
      class="q-mb-sm"
      dense
      color="primary"
      rounded
      @click="handleAdd"
    />
    <category-table
      @edit="handleEdit"
      @delete="handleDelete"
      :categories="categories"
    />
  </div>
</template>

<script>
import moment from "moment";
import CategoryForm from "../components/CategoryForm.vue";
import CategoryTable from "../components/CategoryTable.vue";
export default {

```

```

components: { CategoryForm, CategoryTable },
data() {
  return {
    dialog: false,
    updateDoc: null,
    categories: [],
  };
},
mounted() {
  this.getData();
},
methods: {
  close(doc) {
    if (doc?._id) {
    } else {
      doc._id = new Date();
      this.categories.push(doc);
    }
    console.log("close:", doc);
    this.updateDoc = null;
    this.dialog = false;
  },
  handleDelete(id) {
    console.log("delete on :", id);
    let index = this.categories.findIndex((doc) => {
      return doc._id == id;
    });
    this.$delete(this.categories, index);
  },
  handleEdit(doc) {
    console.log("category:", doc);
    this.updateDoc = Object.assign({}, doc);
    this.updateDoc.date = moment(doc.date).format("YYYY-MM-DD");
    this.dialog = true;
  },
  handleAdd() {
    this.dialog = true;
  },
  getData() {
    this.categories.push({
      _id: "01",
      date: moment(new Date(), "DD-MM-YYYY").toDate(),
      name: "Drink",
      description: "soft-drink",
    });
  },
},
};
</script>

```

ការបង្កើត Form Category

->imports/ui/components/CategoryForm.vue

```
<template>
  <div class="q-pt-md q-mr-xl q-ml-xl">
    <q-card class="shadow-6">
      <q-card-section class="text-h5 text-center">
        Category Form
      </q-card-section>
      <q-card-section>
        <div class="fit row justify-evenly">
          <div class="col-xs-12 col-sm-4 q-ma-sm">
            <q-input square outlined v-
model="form.name" label="Name" />
          </div>

          <div class="col-xs-12 col-sm-4 q-ma-sm">
            <q-input
              type="date"
              square
              outlined
              v-model="form.date"
              label="Date "
            />
          </div>
          <div class="col-xs-12 col-sm-8 q-ma-sm">
            <q-input
              type="textarea"
              square
              outlined
              v-model="form.description"
              label="Description"
            />
          </div>
        </div>
        <div class="text-center">
          <q-btn
            :color="updateDoc ? `info` : `primary`"
            :label="updateDoc ? `Update` : `Submit`"
            @click="handleSubmit"
          />
        </div>
      </q-card-section>
    </q-card>
  </div>
</template>

<script>
import moment from "moment";
export default {
  props: {
    updateDoc: {
```

```

    type: Object,
    default: null,
  },
},
data() {
  return {
    form: {
      name: null,
      date: moment(new Date()).format("YYYY-MM-DD"),
      description: null,
    },
  };
},
created() {
  if (this.updateDoc) {
    this.form = this.updateDoc;
  }
},
methods: {
  handleSubmit() {
    this.form.date = moment(this.form.date, "YYYY-MM-DD").toDate();
    console.log("form:", this.form);
    this.$emit("close", this.form);
  },
},
};
</script>

```

ការបង្កើត Table Category

-> imports/ui/components/CategoryTable.vue

```

<template>
  <div>
    <q-markup-table>
      <thead>
        <tr>
          <th class="text-left">No</th>
          <th class="text-left">Name</th>
          <th class="text-left">Date</th>
          <th class="text-left">Description</th>
          <th class="text-center">Actions</th>
        </tr>
      </thead>
      <tbody>
        <tr v-for="(item, index) in categories" :key="index">
          <td class="text-left">{{ index + 1 }}</td>
          <td class="text-left">{{ item.name }}</td>
          <td class="text-left">{{ item.date | toDate }}</td>
          <td class="text-left">{{ item.description }}</td>
          <td class="text-center">

```

```

        <q-btn
          icon="delete"
          dense
          rounded
          color="red"
          flat
          @click="handleDelete(item._id)"
        />
        <q-btn
          icon="edit"
          dense
          rounded
          color="info"
          flat
          @click="handleEdit(item)"
        />
      </td>
    </tr>
  </tbody>
</q-markup-table>
</div>
</template>

<script>
import moment from "moment";
export default {
  props: {
    categories: {
      type: Array,
      default: null,
    },
  },
  filters: {
    toDate(date) {
      return moment(date).format("DD-MM-YYYY");
    },
  },
  methods: {
    handleDelete(id) {
      this.$emit("delete", id);
    },
    handleEdit(doc) {
      this.$emit("edit", doc);
    },
  },
};
</script>

```


២.Item

No	Name	Category	Date	Description	Status	Actions
1	Coca	Drink	27-03-2021	Coca-1.25L	active	 

Item Form

Name

Category

Date 05/21/2021

Description

Status ☒ Active ☐ Inactive

ការបន្ថែម Routes

->imports/ui/routes/index.js

```
...
{
  path: "/item",
  name: "Item",
  component: () => import("../pages/Item.vue"),
},
...
```

ការបន្ថែម Menus

->imports/ui/App.vue

```
...
<q-item clickable v-ripple to="/item">
  <q-item-section avatar>
    <q-icon name="dns" />
  </q-item-section>
  <q-item-section> Item </q-item-section>
</q-item>
...
```

ការបង្កើត Page Item

->imports/ui/pages/Item.vue

```
<template>
  <div>
    <q-dialog
      v-model="dialog"
      persistent
      :maximized="true">
```

```

        transition-show="slide-up"
        transition-hide="slide-down"
      >
        <q-card>
          <q-bar>
            <q-space />
            <q-btn
              dense
              flat
              icon="close"
              color="white"
              class="bg-red shadow-4"
              rounded
              v-close-popup
            >
              <q-tooltip content-class="bg-white text-
primary ">Close</q-tooltip>
            </q-btn>
          </q-bar>
          <item-form :updateDoc="updateDoc" @close="close" />
        </q-card>
      </q-dialog>
      <q-btn
        icon="add"
        class="q-mb-sm"
        dense
        color="primary"
        rounded
        @click="handleAdd"
      />
    <item-
table @edit="handleEdit" @delete="handleDelete" :items="items" />
  </div>
</template>

<script>
import moment from "moment";
import ItemForm from "../components/ItemForm.vue";
import ItemTable from "../components/ItemTable.vue";

export default {
  components: { ItemForm, ItemTable },
  data() {
    return {
      dialog: false,
      updateDoc: null,
      items: [],
    };
  },
  mounted() {
    this.getData();
  }
};

```

```

    },
    methods: {
      close(doc) {
        if (doc?._id) {
        } else {
          doc._id = new Date();
          this.items.push(doc);
        }
        console.log("close:", doc);
        this.updateDoc = null;
        this.dialog = false;
      },
      handleDelete(id) {
        console.log("delete on :", id);
        let index = this.items.findIndex((doc) => {
          return doc._id == id;
        });
        this.$delete(this.items, index);
      },
      handleEdit(doc) {
        console.log("customer:", doc);
        this.updateDoc = Object.assign({}, doc);
        this.updateDoc.date = moment(doc.date).format("YYYY-MM-DD");
        this.dialog = true;
      },
      handleAdd() {
        this.dialog = true;
      },
      getData() {
        this.items.push({
          _id: "01",
          name: "Coca",
          categoryName: "Drink",
          categoryId: "01",
          date: moment("27-3-2021", "DD-MM-YYYY").toDate(),
          description: "Coca-1.25L",
          status: "active",
        });
      },
    },
  },
};
</script>

```

ការបង្កើត Component Form Item

-> imports/ui/components/ItemForm.vue

```

<template>
  <div class="q-pt-md q-mr-xl q-ml-xl">
    <q-card class="shadow-6">

```

```

    <q-card-section class="text-h5 text-center"> Item Form </q-
card-section>
    <q-card-section>
      <div class="fit row justify-evenly">
        <div class="col-xs-12 col-sm-4 q-ma-sm">
          <q-input square outlined v-
model="form.name" label="Name" />
        </div>
        <div class="col-xs-12 col-sm-4 q-ma-sm">
          <q-select
            square
            outlined
            v-model="form.categoryId"
            label="Category"
            :options="categoryOpts"
            emit-value
            map-options
            option-label="name"
            option-value="_id"
          />
        </div>

        <div class="col-xs-12 col-sm-4 q-ma-sm">
          <q-input
            type="date"
            square
            outlined
            v-model="form.date"
            label="Date "
          />
        </div>
        <div class="col-xs-12 col-sm-4 q-ma-sm">
          <q-input
            type="textarea"
            square
            outlined
            v-model="form.description"
            label="Description"
            autogrow
          />
        </div>
        <div class="col-xs-12 col-sm-4 q-ma-sm">
          <fieldset>
            <label class="text-grey text-subtitle1">Status</label>
            <q-radio v-
model="form.status" val="active" label="Active" />
            <q-radio v-
model="form.status" val="inactive" label="Inactive" />
          </fieldset>
        </div>
      </div>
    </div>
  </div>

```

```

        <div class="text-center">
            <q-
btn color="primary" label="Submit" @click="handleSubmit" />
        </div>
    </q-card-section>
</q-card>
</div>
</template>

<script>
import moment from "moment";
export default {
  props: {
    updateDoc: {
      type: Object,
      default: null,
    },
  },
  data() {
    return {
      form: {
        name: null,
        categoryId: null,
        date: moment(new Date()).format("YYYY-MM-DD"),
        description: null,
        status: "active",
      },
      categoryOpts: [],
    };
  },
  created() {
    this.getCategory();
    if (this.updateDoc) {
      this.form = this.updateDoc;
    }
  },
  methods: {
    getCategory() {
      this.categoryOpts = [
        {
          _id: "01",
          name: "Drink",
        },
        {
          _id: "02",
          name: "Food",
        },
      ];
    },
  },
};

```

```

    handleSubmit() {
      this.form.date = moment(this.form.date, "YYYY-MM-DD").toDate();
      console.log("form:", this.form);
      this.form.categoryName = this.form.categoryId;
      this.$emit("close", this.form);
    },
  },
};
</script>

```

ការបង្កើត Table Item

->imports/ui/components/ItemTable.vue

```

<template>
  <div>
    <q-markup-table>
      <thead>
        <tr>
          <th class="text-left">No</th>
          <th class="text-left">Name</th>
          <th class="text-left">Category</th>
          <th class="text-left">Date</th>
          <th class="text-left">Description</th>
          <th class="text-left">Status</th>
          <th class="text-center">Actions</th>
        </tr>
      </thead>
      <tbody>
        <tr v-for="(item, index) in items" :key="index">
          <td class="text-left">{{ index + 1 }}</td>
          <td class="text-left">{{ item.name }}</td>
          <td class="text-left">{{ item.categoryName }}</td>
          <td class="text-left">{{ item.date | toDate }}</td>
          <td class="text-left">{{ item.description }}</td>
          <td class="text-left">{{ item.status }}</td>
          <td class="text-center">
            <q-btn
              icon="delete"
              dense
              rounded
              color="red"
              flat
              @click="handleDelete(item._id)"
            />
            <q-btn
              icon="edit"
              dense
              rounded
              color="info"

```

```

        flat
        @click="handleEdit(item)"
      />
    </td>
  </tr>
</tbody>
</q-markup-table>
</div>
</template>

<script>
import moment from "moment";
export default {
  props: {
    items: {
      type: Array,
      default: null,
    },
  },
  filters: {
    toDate(date) {
      return moment(date).format("DD-MM-YYYY");
    },
  },
  methods: {
    handleDelete(id) {
      this.$emit("delete", id);
    },
    handleEdit(doc) {
      this.$emit("edit", doc);
    },
  },
};
</script>

```

៥.Supplier

Master IT Shop						
Supplier						
+						
No	Company Name	Owner Name	Phone	Address	Status	Actions
1	Cambodia Soft-Drink	Chetra	010202022	Battambang	active	 

ការបន្ថែម Routes

->imports/ui/routes/index.js

```
...
{
  path: "/supplier",
  name: "Supplier",
  component: () => import("../pages/Supplier.vue"),
},
...
```

ការបន្ថែម Menus

->imports/ui/App.vue

```
...
<q-item clickable v-ripple to="/supplier">
  <q-item-section avatar>
    <q-icon name="maps_home_work" />
  </q-item-section>
  <q-item-section> Supplier </q-item-section>
</q-item>
...
```

ការបង្កើត Page Supplier

->imports/ui/pages/Supplier.vue

```
<template>
  <div>
    <q-dialog
      v-model="dialog"
      persistent
      :maximized="true"
      transition-show="slide-up">
```



```

        transition-hide="slide-down"
      >
        <q-card>
          <q-bar>
            <q-space />
            <q-btn
              dense
              flat
              icon="close"
              color="white"
              class="bg-red shadow-4"
              rounded
              v-close-popup
            >
              <q-tooltip content-class="bg-white text-
primary ">Close</q-tooltip>
            </q-btn>
          </q-bar>
          <supplier-form :updateDoc="updateDoc" @close="close" />
        </q-card>
      </q-dialog>
      <q-btn
        icon="add"
        class="q-mb-sm"
        dense
        color="primary"
        rounded
        @click="handleAdd"
      />
      <supplier-table
        @edit="handleEdit"
        @delete="handleDelete"
        :suppliers="suppliers"
      />
    </div>
  </template>

  <script>
  import SupplierForm from "../components/SupplierForm.vue";
  import SupplierTable from "../components/SupplierTable.vue";
  export default {
    components: {
      SupplierForm,
      SupplierTable,
    },
    data() {
      return {
        dialog: false,
        updateDoc: null,
        suppliers: [],

```

```

    };
  },
  mounted() {
    this.getData();
  },
  methods: {
    close(doc) {
      if (doc?._id) {
      } else {
        doc._id = new Date();
        this.suppliers.push(doc);
      }
      console.log("close:", doc);
      this.updateDoc = null;
      this.dialog = false;
    },
    handleDelete(id) {
      console.log("delete on :", id);
      let index = this.suppliers.findIndex((doc) => {
        return doc._id == id;
      });
      this.$delete(this.suppliers, index);
    },
    handleEdit(doc) {
      console.log("supplier:", doc);
      this.updateDoc = Object.assign({}, doc);
      this.dialog = true;
    },
    handleAdd() {
      this.dialog = true;
    },
    getData() {
      this.suppliers = [
        {
          _id: "01",
          company: "Cambodia Soft-Drink",
          ownerName: "Chetra",
          phone: "010202022",
          address: "Battambang",
          status: "active",
        },
      ];
    },
  },
};
</script>

```

ការបង្កើត Form Supplier

->imports/ui/components/SupplierForm.vue

```

<template>
  <div class="q-pr-xl q-pl-xl q-pt-md q-mr-xl q-ml-xl">
    <q-card class="shadow-6">
      <q-card-section class="text-h5 text-center">
        Supplier Form
      </q-card-section>
      <q-card-section>
        <div class="fit row justify-evenly">
          <div class="col-xs-12 col-sm-4 q-ma-sm">
            <q-input
              square
              outlined
              v-model="form.company"
              label="Company Name"
            />
          </div>
          <div class="col-xs-12 col-sm-4 q-ma-sm">
            <q-input
              square
              outlined
              v-model="form.ownerName"
              label="Owner Name"
            />
          </div>
          <div class="col-xs-12 col-sm-4 q-ma-sm">
            <q-input
              square
              outlined
              v-model="form.phone"
              label="Phone Number"
            />
          </div>
          <div class="col-xs-12 col-sm-4 q-ma-sm">
            <q-input
              square
              outlined
              v-model="form.address"
              label="Address"
              autogrow
              hint="Country/Province/District/Village"
            />
          </div>
          <div class="col-xs-12 col-sm-4 q-ma-sm">
            <fieldset>
              <label class="text-grey text-subtitle1">Status</label>
              <q-radio v-
model="form.status" val="active" label="Active" />
              <q-radio v-
model="form.status" val="inactive" label="Inactive" />
            </fieldset>
          </div>
        </div>
      </q-card-section>
    </q-card>
  </div>

```

```

    </div>
    <div class="col-xs-12 col-sm-4 q-ma-sm"></div>
    <div class="col-xs-12 col-sm-4 q-ma-sm">
      <q-btn
        :color="updateDoc ? `info` : `primary`"
        :label="updateDoc ? `Update` : `Submit`"
        @click="handleSubmit"
      />
    </div>
    <div class="col-xs-12 col-sm-4 q-ma-sm"></div>
  </div>
</q-card-section>
</q-card>
</div>
</template>

<script>
export default {
  props: {
    updateDoc: {
      type: Object,
      default: null,
    },
  },
  data() {
    return {
      form: {
        company: null,
        ownerName: null,
        phone: null,
        address: null,
        status: "active",
      },
    };
  },
  created() {
    if (this.updateDoc) {
      this.form = this.updateDoc;
    }
  },
  methods: {
    handleSubmit() {
      this.$emit("close", this.form);
    },
  },
};
</script>

```

ការបង្កើត Table Supplier

```
->imports/ui/components/SupplierTable.vue
```

```
<template>
  <div>
    <q-markup-table>
      <thead>
        <tr>
          <th class="text-left">No</th>
          <th class="text-left">Company Name</th>
          <th class="text-left">Owner Name</th>
          <th class="text-left">Phone</th>
          <th class="text-left">Address</th>
          <th class="text-left">Status</th>
          <th class="text-center">Actions</th>
        </tr>
      </thead>
      <tbody>
        <tr v-for="(supplier, index) in suppliers" :key="index">
          <td class="text-left">{{ index + 1 }}</td>
          <td class="text-left">{{ supplier.company }}</td>
          <td class="text-left">{{ supplier.ownerName }}</td>
          <td class="text-left">{{ supplier.phone }}</td>
          <td class="text-left">{{ supplier.address }}</td>
          <td class="text-left">{{ supplier.status }}</td>
          <td class="text-center">
            <q-btn
              icon="delete"
              dense
              rounded
              color="red"
              flat
              @click="handleDelete(supplier._id)"
            />
            <q-btn
              icon="edit"
              dense
              rounded
              color="info"
              flat
              @click="handleEdit(supplier)"
            />
          </td>
        </tr>
      </tbody>
    </q-markup-table>
  </div>
</template>

<script>
export default {
  props: {
```

```

suppliers: {
  type: Array,
  default: null,
},
},
methods: {
  handleDelete(id) {
    this.$emit("delete", id);
  },
  handleEdit(doc) {
    this.$emit("edit", doc);
  },
},
};
</script>

```

៧. Purchase

Master IT Shop

purchase

No	Name	Cost	Price	Quantity	Date	Status
1	Pizza	4000	5000	13	29-06-2021	active
2	Coca	1200	2500	18	29-06-2021	active
3	Spy	4500	6000	12	29-06-2021	active

Purchase Form

Supplier:

Cost:

Quantity:

Status: ☒ Active ☐ Inactive

Item:

Price:

Date: 06/29/2021

SUBMIT

ការបង្កើត Routes

-> imports/ui/routes/index.js

```

...
{
  path: "/purchase",
  name: "Purchase",
  component: () => import("../pages/Purchase.vue"),
},
...

```

ការបន្ថែម Menus

->imports/ui/App.vue

```
...
    <q-item class="item" clickable v-ripple to="/purchase">
      <q-item-section avatar>
        <q-icon name="inventory" />
      </q-item-section>
      <q-item-section> Purchase </q-item-section>
    </q-item>
  ...
```

ការបង្កើត Page Purchase

->imports/ui/pages/Purchase.vue

```
<template>
  <div>
    <q-dialog
      v-model="dialog"
      persistent
      :maximized="true"
      transition-show="slide-up"
      transition-hide="slide-down"
    >
      <q-card>
        <q-bar>
          <q-space />
          <q-btn
            dense
            flat
            icon="close"
            color="white"
            class="bg-red shadow-4"
            rounded
            v-close-popup
          >
            <q-tooltip content-class="bg-white text-
primary ">Close</q-tooltip>
          </q-btn>
        </q-bar>
        <purchase-form @close="close" />
      </q-card>
    </q-dialog>
    <q-btn
      icon="add"
      class="q-mb-sm"
      dense
```

```

        color="primary"
        rounded
        @click="handleAdd"
      />
    <purchase-table :items="items" />
  </div>
</template>

<script>
import PurchaseForm from "../components/PurchaseForm.vue";
import PurchaseTable from "../components/PurchaseTable.vue";
export default {
  components: { PurchaseTable, PurchaseForm },
  data() {
    return {
      dialog: false,
      items: [],
    };
  },
  mounted() {
    this.getData();
  },
  methods: {
    handleAdd() {
      this.dialog = true;
    },
    close(doc) {
      if (doc?._id) {
      } else {
        doc._id = new Date();
        this.items.push(doc);
      }
      console.log("close:", doc);
      this.dialog = false;
    },
    getData() {
      this.items = [
        {
          _id: "01",
          supplierId: "YA679h36eWhwSW4pK",
          itemId: "01",
          name: "Pizza",
          cost: 4000,
          price: 5000,
          qty: 13,
          date: new Date(),
          status: "active",
        },
        {
          _id: "02",

```



```

        supplierId: "GRx83b78chGwCftKa",
        itemId: "02",
        name: "Coca",
        cost: 1200,
        price: 2500,
        qty: 18,
        date: new Date(),
        status: "active",
      },
    ],
  },
},
};
</script>

```

ការបង្កើត Form Purchase

-> imports/ui/components/PurchaseForm.vue

```

<template>
  <div class="q-pt-md q-mr-xl q-ml-xl">
    <q-card class="shadow-6">
      <q-card-section class="text-h5 text-center">
        Purchase Form
      </q-card-section>
      <q-card-section>
        <div class="fit row justify-evenly">
          <div class="col-xs-12 col-sm-4 q-ma-sm">
            <q-select
              square
              outlined
              v-model="form.supplierId"
              label="Supplier"
              :options="supplierOpts"
              emit-value
              map-options
              option-label="company"
              option-value="_id"
            />
          </div>
          <div class="col-xs-12 col-sm-4 q-ma-sm">
            <q-select
              square
              outlined
              v-model="form.itemId"
              label="Item"
              :options="itemOpts"
              emit-value
              map-options
              option-label="name"
            />
          </div>
        </div>
      </q-card-section>
    </q-card>
  </div>
</template>

```

```

        option-value="_id"
      />
    </div>

    <div class="col-xs-12 col-sm-4 q-ma-sm">
      <q-input
        square
        outlined
        v-model.number="form.cost"
        label="Cost "
        type="number"
      />
    </div>
    <div class="col-xs-12 col-sm-4 q-ma-sm">
      <q-input
        square
        outlined
        v-model.number="form.price"
        label="Price"
        type="number"
      />
    </div>
    <div class="col-xs-12 col-sm-4 q-ma-sm">
      <q-input
        square
        outlined
        v-model.number="form.qty"
        label="Quantity"
        type="number"
      />
    </div>
    <div class="col-xs-12 col-sm-4 q-ma-sm">
      <q-input
        type="date"
        square
        outlined
        v-model="form.date"
        label="Date "
      />
    </div>

    <div class="col-xs-12 col-sm-4 q-ma-sm">
      <fieldset>
        <label class="text-grey text-subtitle1">Status</label>
        <q-radio v-
model="form.status" val="active" label="Active" />
        <q-radio v-
model="form.status" val="inactive" label="Inactive" />
      </fieldset>
    </div>
    <div class="col-xs-12 col-sm-4 q-ma-sm"></div>

```

```

        <div class="col-xs-12 col-sm-4 q-ma-sm">
            <q-
btn color="primary" label="Submit" @click="handleSubmit" />
        </div>
        <div class="col-xs-12 col-sm-4 q-ma-sm"></div>
    </div>
</q-card-section>
</q-card>
</div>
</template>

<script>
import { Notify } from "quasar";
import moment from "moment";
export default {
  data() {
    return {
      form: {
        supplierId: null,
        itemId: null,
        cost: null,
        price: null,
        qty: null,
        date: moment(new Date()).format("YYYY-MM-DD"),
        status: "active",
      },
      itemOpts: [],
      supplierOpts: [],
    };
  },
  mounted() {
    this.getItem();
    this.getSupplier();
  },
  methods: {
    handleSubmit() {
      let valid = false;
      for (const doc in this.form) {
        if (this.form[doc] == null) {
          valid = true;
        }
      }
      if (!valid) {
        // code add purchase
        // -----set item name-----
        let index = this.itemOpts.findIndex((doc) => {
          return doc._id == this.form.itemId;
        });
        this.form.name = this.itemOpts[index].name;
        // -----

```

```

        this.$emit("close", this.form);
    } else {
        Notify.create({
            message: "Please fill the form ",
            color: "negative",
            type: "negative",
            position: "top",
        });
    }
},
getItem() {
    this.itemOpts = [
        {
            _id: "01",
            name: "Coca",
            categoryName: "Drink",
            categoryId: "01",
            date: moment("27-3-2021", "DD-MM-YYYY").toDate(),
            description: "Coca-1.25L",
            status: "active",
        },
        {
            _id: "02",
            name: "Spy",
            categoryName: "Drink",
            categoryId: "01",
            date: moment("27-3-2021", "DD-MM-YYYY").toDate(),
            description: "Coca-1.25L",
            status: "active",
        },
    ];
},
getSupplier() {
    this.supplierOpts = [
        {
            _id: "01",
            company: "Cambodia Soft-Drink",
            ownerName: "Chetra",
            phone: "010202022",
            address: "Battambang",
            status: "active",
        },
    ];
},
};
</script>

```

ការបង្កើត Table Purchase

->imports/ui/components/PurchaseTable.vue

```

<template>
  <div>
    <q-markup-table>
      <thead>
        <tr>
          <th class="text-left">No</th>
          <th class="text-left">Name</th>
          <th class="text-left">Cost</th>
          <th class="text-left">Price</th>
          <th class="text-left">Quantity</th>
          <th class="text-left">Date</th>
          <th class="text-left">Status</th>
        </tr>
      </thead>
      <tbody>
        <tr v-for="(item, index) in items" :key="index">
          <td class="text-left">{{ index + 1 }}</td>
          <td class="text-left">{{ item.name }}</td>
          <td class="text-left">{{ item.cost }}</td>
          <td class="text-left">{{ item.price }}</td>
          <td class="text-left">{{ item.qty }}</td>
          <td class="text-left">{{ item.date | toDate }}</td>
          <td class="text-left">{{ item.status }}</td>
        </tr>
      </tbody>
    </q-markup-table>
  </div>
</template>
<script>
import moment from "moment";
export default {
  props: {
    items: {
      type: Array,
      default: null,
    },
  },
  filters: {
    toDate(date) {
      return moment(date).format("DD-MM-YYYY");
    },
  },
};
</script>

```

ជ. Sale

ការបន្ថែម Routes

->imports/ui/routes/index.js

```
...
{
  path: "/sale",
  component: () => import("../pages/Sale.vue"),
},
...
```

ការបន្ថែម Menus

->imports/ui/App.vue

```
...
<q-separator />
<q-item class="item" clickable v-ripple to="/sale">
  <q-item-section avatar>
    <q-icon name="chrome_reader_mode" />
  </q-item-section>
  <q-item-section> Sale </q-item-section>
</q-item>
...
```

ការបង្កើត Page Sale

->imports/ui/pages/Sale.vue

```
<template>
  <div class="q-pt-md q-mr-xl q-ml-xl">
    <form @submit.prevent="handleSubmit">
      <q-card class="shadow-6">
        <q-card-section class="text-h5 text-center">Sale Form </q-
card-section>
```

```

<q-card-section>
  <div class="row items-center">
    <div class="col-md-8 col-xs-12">
      <div class="col-xs-12 col-sm-4 q-ma-sm">
        <q-input
          square
          outlined
          type="number"
          readonly
          v-model.number="form.invoiceNumber"
          label="Invoice Number"
        />
      </div>
      <div class="col-xs-12 col-sm-4 q-ma-sm">
        <q-input
          square
          outlined
          readonly
          type="date"
          v-model="form.date"
          label="Date"
        />
      </div>
      <div class="col-xs-12 col-sm-4 q-ma-sm">
        <q-select
          square
          outlined
          v-model="form.customerId"
          :options="customerOpts"
          label="Customer"
          map-options
          emit-value
          option-label="name"
          option-value="_id"
          required
        />
      </div>
      <div class="col-xs-12 col-sm-4 q-ma-sm">
        <q-select
          square
          outlined
          v-model="item"
          :options="itemOpts"
          label="Items"
          emit-value
          map-options
          option-label="name"
          @input="handleChangeItem"
        />
      </div>
    </div>
  </div>

```

```

    </div>
    <div class="col-md-4 col-xs-12">
      <q-card flat class="q-pa-xl q-pr-xl">
        <div class="text-h5 text-
center">Total : {{ total() }}</div>
      </q-card>
    </div>
  </div>

  <q-markup-table>
    <thead>
      <tr>
        <th class="text-left">No</th>
        <th class="text-left">Name</th>
        <th class="text-left">Price</th>
        <th class="text-left">Qty</th>
        <th class="text-left">Amount</th>
        <th class="text-left">Action</th>
      </tr>
    </thead>
    <tbody>
      <tr v-for="(item, index) in form.items" :key="index">
        <td class="text-left">{{ index + 1 }}</td>
        <td class="text-left">{{ item.name }}</td>
        <td class="text-left">{{ item.price }}</td>
        <td class="text-left" style="width: 150px">
          <q-input type="number" v-model.number="item.qty" />
        </td>
        <td class="text-
left">{{ item.price * item.qty }}</td>
        <td class="text-left">
          <q-btn
            color="red"
            icon="delete"
            dense
            flat
            @click="handleRemove(index)"
          />
        </td>
      </tr>
    </tbody>
  </q-markup-table>
  <div
    class="text-center q-mt-lg"
    v-if="form.items.length != 0 && form.customerId"
  >
    <q-btn color="primary" label="Submit" type="submit" />
  </div>
</q-card-section>
</q-card>
</form>

```



```

</div>
</template>

<script>
import moment from "moment";
export default {
  data() {
    return {
      item: null,
      form: {
        invoiceNumber: null,
        date: moment(new Date()).format("YYYY-MM-DD"),
        customerId: null,
        items: [],
      },
      itemOpts: [],
      customerOpts: [],
    };
  },
  mounted() {
    this.getInvoiceNumber();
    this.getCustomer();
    this.getPurchase();
  },
  methods: {
    handleSubmit() {
      let doc = {
        invoiceNumber: this.form.invoiceNumber,
        customerId: this.form.customerId,
        date: moment(this.form.date, "YYYY-MM-DD").toDate(),
        items: [],
      };
      this.form.items.forEach((it) => {
        doc.items.push({
          _id: it._id,
          price: it.price,
          qty: it.qty,
        });
      });

      console.log("document: ", doc);
      this.clearForm();
    },
    handleChangeItem(val) {
      let found = this.form.items.findIndex((doc) => {
        return doc._id == val._id;
      });
      if (found != -1) {
        this.form.items[found].qty++;
      } else {

```

```

        this.form.items.push({
          _id: val.itemId,
          name: val.name,
          price: val.price,
          qty: 1,
        });
      },
      handleRemove(index) {
        this.$delete(this.form.items, index);
      },
      total() {
        let total = 0;
        this.form.items.forEach((it) => {
          total += it.price * it.qty;
        });
        return total;
      },
      clearForm() {
        this.form = {
          invoiceNumber: null,
          date: moment(new Date()).format("YYYY-MM-DD"),
          customerId: null,
          items: [],
        };
        this.item = null;
        this.getInvoiceNumber();
      },
      getInvoiceNumber() {
        this.form.invoiceNumber = 1;
      },
      getPurchase() {
        this.itemOpts = [
          {
            _id: "01",
            itemId: "01",
            name: "Coca",
            price: 5500,
            qty: 15,
          },
          {
            _id: "02",
            itemId: "02",
            name: "Spy",
            price: 6000,
            qty: 12,
          },
        ];
      },
      getCustomer() {
        this.customerOpts = [

```

```

    {
      _id: "01",
      name: "Sopheak Chean",
    },
    {
      _id: "02",
      name: "Leakena",
    },
  ],
},
},
};
</script>

```

៧៥.Component Query By Date

Search By Date

ការបង្កើត Form Query By Date

->imports/ui/components/FormSearchByDate.vue

```

<template>
  <fieldset>
    <legend>Search By Date</legend>
    <div class="row justify-center items-center">
      <div class="col-md-4 col-xs-12 q-pa-sm">
        <q-input outlined dense v-model="form.fromDate">
          <template v-slot:append>
            <q-icon name="event" class="cursor-pointer">
              <q-popup-proxy
                ref="qDateProxy"
                transition-show="scale"
                transition-hide="scale"
              >
                <q-date mask="DD-MM-YYYY" v-model="form.fromDate">
                  <div class="row items-center justify-end">
                    <q-btn v-close-
                      popup label="Close" color="primary" flat />
                  </div>
                </q-date>
              </q-popup-proxy>
            </q-icon>
          </template>
        </q-input>
      </div>
      <div class="col-md-4 col-xs-12 q-pa-sm">
        <q-input outlined dense v-model="form.toDate">

```

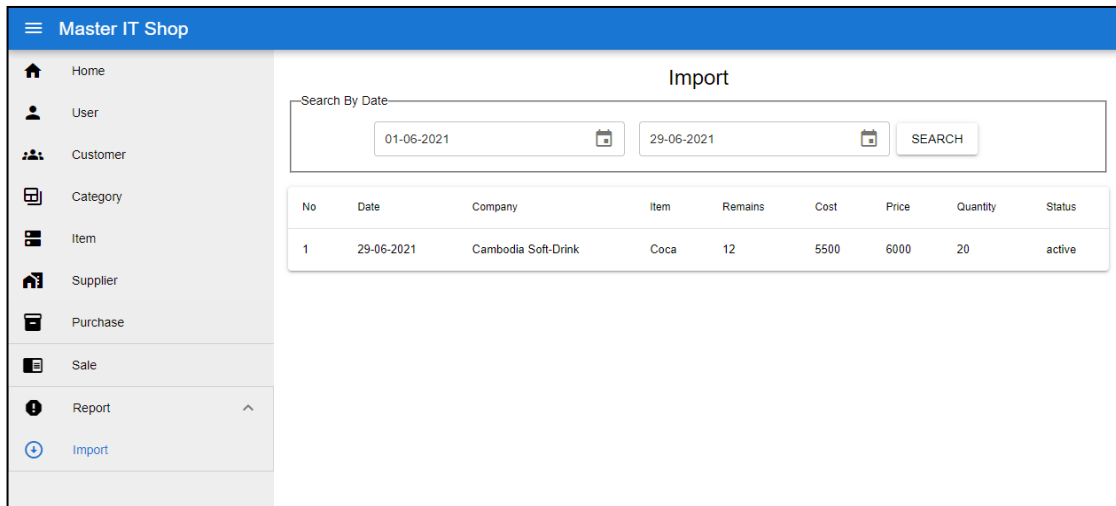
```

    <template v-slot:append>
      <q-icon name="event" class="cursor-pointer">
        <q-popup-proxy
          ref="qDateProxy"
          transition-show="scale"
          transition-hide="scale"
        >
          <q-date mask="DD-MM-YYYY" v-model="form.toDate">
            <div class="row items-center justify-end">
              <q-btn v-close-
popup label="Close" color="primary" flat />
            </div>
          </q-date>
        </q-popup-proxy>
      </q-icon>
    </template>
  </q-input>
</div>
<div class="col-md-2 col-xs-12">
  <q-btn label="Search" @click="findByDate()" />
</div>
</div>
</fieldset>
</template>

<script>
import moment from "moment";
export default {
  data() {
    return {
      form: {
        fromDate: moment(new Date()).startOf("month").format("DD-MM-
YYYY"),
        toDate: moment(new Date()).format("DD-MM-YYYY"),
      },
    };
  },
  mounted() {
    this.findByDate();
  },
  methods: {
    findByDate() {
      this.$emit("searchByDate", this.form);
    },
  },
};
</script>

```

៣.Import



ការបន្ថែម Routes

->imports/ui/routes/index.js

```
...
{
  path: "/import",
  name: "Import",
  component: () => import("../pages/Import.vue"),
},
...
```

ការបន្ថែម Menus

->imports/ui/App.vue

```
...
<q-list bordered class="item">
  <q-expansion-item icon="report" label="Report">
    <q-item class="item" clickable v-ripple to="import">
      <q-item-section avatar>
        <q-icon name="arrow_circle_down" />
      </q-item-section>
      <q-item-section> Import </q-item-section>
    </q-item>
  </q-expansion-item>
</q-list>
...
```

ការបង្កើត Page Import

->imports/ui/pages/Import.vue

```
<template>
```

```

<div>
  <form-search-by-date @searchByDate="searchByDate" />
  <div class="q-mt-md">
    <q-markup-table>
      <thead>
        <tr>
          <th class="text-left">No</th>
          <th class="text-left">Date</th>
          <th class="text-left">Company</th>
          <th class="text-left">Item</th>
          <th class="text-left">Remains</th>
          <th class="text-left">Cost</th>
          <th class="text-left">Price</th>
          <th class="text-left">Quantity</th>
          <th class="text-left">Status</th>
        </tr>
      </thead>
      <tbody>
        <tr v-for="(item, index) in items" :key="index">
          <td class="text-left">{{ index + 1 }}</td>
          <td class="text-left">{{ item.date | toDate }}</td>
          <td class="text-left">{{ item.company }}</td>
          <td class="text-left">{{ item.name }}</td>
          <td class="text-left">{{ item.remains }}</td>
          <td class="text-left">{{ item.cost }}</td>
          <td class="text-left">{{ item.price }}</td>
          <td class="text-left">{{ item.qty }}</td>
          <td class="text-left">{{ item.status }}</td>
        </tr>
      </tbody>
    </q-markup-table>
  </div>
</div>
</template>

<script>
import moment from "moment";
import FormSearchByDate from "../components/FormSearchByDate.vue";

export default {
  components: { FormSearchByDate },
  data() {
    return {
      items: [],
    };
  },
  filters: {
    toDate(date) {
      return moment(date).format("DD-MM-YYYY");
    },
  },
};

```

```

methods: {
  searchByDate(date) {
    console.log("date", date);
    this.items = [
      {
        _id: "01",
        date: new Date(),
        company: "Cambodia Soft-Drink",
        name: "Coca",
        remains: 12,
        cost: 5500,
        price: 6000,
        qty: 20,
        status: "active",
      },
    ];
  },
},
};
</script>

```

៨.Export

Header						
Home	Search By Date 20-05-2021 21-05-2021 SEARCH					
Customer						
Category						
Item						
Inventory						
Order						
Import						
Export						

No	Date	Customer	Item	Price	Quantity
0	02-04-2021	Monica	Coca	6000	2
			Spy	6000	3

ការបង្កើត Routes

-> imports/ui/routes/index.js

```

...
{
  path: "/export",
  name: "Export",
  component: () => import("../pages/Export.vue"),
},
...

```

ការបង្កើត Menus

-> imports/ui/App.vue

```

...
<q-list bordered class="item">
  <q-expansion-item icon="report" label="Report">
    <q-item class="item" clickable v-ripple to="import">

```

```

        <q-item-section avatar>
          <q-icon name="arrow_circle_down" />
        </q-item-section>
        <q-item-section> Import </q-item-section>
      </q-item>
      <q-item class="item" clickable v-ripple to="export">
        <q-item-section avatar>
          <q-icon name="arrow_circle_up" />
        </q-item-section>
        <q-item-section> Export </q-item-section>
      </q-item>
    </q-expansion-item>
  </q-list>

```

ការបង្កើត Page Export

->imports/ui/pages/Export.vue

```

<template>
  <div>
    <form-search-by-date @searchByDate="searchByDate" />
    <div class="q-mt-md">
      <q-markup-table>
        <thead>
          <tr>
            <th class="text-left">No</th>
            <th class="text-left">Date</th>
            <th class="text-left">Customer</th>
            <th class="text-left">Item</th>
            <th class="text-left">Price</th>
            <th class="text-left">Quantity</th>
          </tr>
        </thead>
        <tbody>
          <template v-for="(item, index) in items">
            <tr :key="index">
              <td class="text-left">{{ index }}</td>
              <td class="text-left">{{ item.date | toDate }}</td>
              <td class="text-left">{{ item.customerName }}</td>
              <td class="text-left">{{ item.items[0].name }}</td>
              <td class="text-left">{{ item.items[0].price }}</td>
              <td class="text-left">{{ item.items[0].qty }}</td>
            </tr>
            <template v-for="(it, idx) in item.items">
              <tr :key="idx + `it` + index" v-if="idx != 0">
                <td colspan="3"></td>
                <td class="text-left">{{ it.name }}</td>
                <td class="text-left">{{ it.price }}</td>
                <td class="text-left">{{ it.qty }}</td>
              </tr>
            </template>
          </template>
        </tbody>
      </q-markup-table>
    </div>
  </div>
</template>

```



```

        </tr>
      </template>
    </template>
  </tbody>
</q-markup-table>
</div>
</div>
</template>

<script>
import moment from "moment";
import FormSearchByDate from "../components/FormSearchByDate.vue";
export default {
  components: { FormSearchByDate },
  data() {
    return {
      items: [],
    };
  },
  filters: {
    toDate(date) {
      return moment(date).format("DD-MM-YYYY");
    },
  },
  methods: {
    searchByDate(date) {
      console.log("date", date);
      this.items = [
        {
          _id: "01",
          date: new Date(),
          customerName: "Biona",
          items: [
            {
              _id: "01",
              name: "Coca",
              price: 6000,
              qty: 2,
            },
            {
              _id: "02",
              name: "Spy",
              price: 6000,
              qty: 3,
            },
          ],
        },
      ];
    },
  },
};

```

```
};
</script>
```

១.៥ ផ្លាស់ប្តូរ Dashboard

public\img\logo.png



->imports/ui/pages/Home.vue

```
<template>
  <div>
    <q-dialog
      persistent
      :maximized="true"
      transition-show="slide-up"
      transition-hide="slide-down"
      v-model="dialog"
    >
      <q-card>
        <q-bar>
          <q-space />
          <q-btn
            dense
            flat
            icon="close"
            color="white"
            class="bg-red shadow-4"
            rounded
            v-close-popup
          >
            <q-tooltip content-class="bg-white text-primary ">Close</q-
tooltip>
          </q-btn>
        </q-bar>
        <purchase-form @close="close" />
      </q-card>
    </q-dialog>
```

```

    <div class="absolute-full text-subtitle2 flex flex-center q-mb-xl q-
pb-xl">
      <div class="text-center items-center">
        <q-img src="img/logo.png" width="720px" />
      </div>
      <div class="full-width row wrap justify-center items-
start content-start">
        <div class="q-ma-lg" v-
for="(list, index) in lists" :key="index">
          <q-btn
            class="btn"
            :color="list.background"
            @click="handleEnter(list.path)"
          >
            <q-icon class="col-
12" left :name="list.icon" size="10rem" />
            <div class="col-12 text-h4">{{ list.label }}</div>
          </q-btn>
        </div>
      </div>
    </div>
  </template>

<script>
import PurchaseForm from "../components/PurchaseForm.vue";
export default {
  components: { PurchaseForm },
  data() {
    return {
      dialog: false,
      lists: [
        {
          path: "user",
          icon: "person",
          label: "User",
          background: "blue-6",
        },
        {
          path: "customer",
          icon: "groups",
          label: "Customer",
          background: "red-6",
        },
        {
          path: "supplier",
          icon: "maps_home_work",
          label: "Supplier",
          background: "amber-6",
        },
      ],
    };
  },
};

```

```

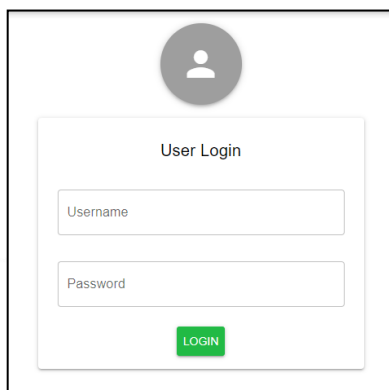
    {
      path: "category",
      icon: "backup_table",
      label: "Category",
      background: "green-6",
    },
    {
      path: "item",
      icon: "dns",
      label: "Item",
      background: "blue-6",
    },
    {
      path: "purchase",
      icon: "inventory",
      label: "Purchase",
      background: "red-6",
    },
    {
      path: "sale",
      icon: "add_shopping_cart",
      label: "Quick Sale",
      background: "amber-6",
    },
    {
      path: false,
      icon: "arrow_circle_down",
      label: "Quick Purchase",
      background: "green-6",
    },
  ],
};
},
methods: {
  handleEnter(path) {
    if (path) {
      this.$router.push(path);
    } else {
      this.dialog = true;
    }
  },
  close() {
    this.dialog = false;
  },
},
};
</script>
<style>
.btn :hover {
  transform: scale(0.98);
  transition: transform 0.4s;
}

```

```
}  
.btn {  
  width: 350px;  
  height: 250px;  
  font-size: 20px;  
}  
.slide-fade-enter-active {  
  transition: all 1.5s ease;  
}  
.slide-fade-leave-active {  
  transition: all 1.5s cubic-bezier(1, 0.5, 0.8, 1);  
}  
.slide-fade-enter, .slide-fade-leave-to  
/* .slide-fade-leave-active below version 2.1.8 */ {  
  transform: translateX(10px);  
  opacity: 0;  
}  
</style>
```



៧.Login



A mockup of a user login form. At the top is a grey circular avatar icon with a white person silhouette. Below it is a white card with a grey border. The card has a title 'User Login' in bold. It contains two input fields: 'Username' and 'Password'. Below the input fields is a green button with the text 'LOGIN' in white.

ការបន្ថែម Routes

->imports/ui/routes/index.js

```
...
{
  path: "/login",
  component: () => import("../pages/Login.vue"),
},
...
```

ការបង្កើត Page Login

->imports/ui/pages/Login.vue

```
<template>
  <div>
    <div class="row justify-center q-mt-xl">
      <div class="col-12 text-center q-mb-md">
        <q-avatar
          size="100px"
          color="grey"
          text-color="white"
          icon="person"
          class="shadow-3"
        />
      </div>
    </div>
    <q-card class="col-12 q-pa-sm" style="width: 400px">
      <q-card-section class="text-center">
        <label class="text-center text-h6">User Login</label>
      </q-card-section>
      <q-card-section>
        <q-input outlined v-
model="form.username" label="Username" />
      </q-card-section>
      <q-card-section>
        <q-input
```

```

        outlined
        v-model="form.password"
        type="password"
        label="Password"
      />
    </q-card-section>
    <q-card-actions align="center">
      <q-
btn color="positive" label="Login" @click="handleLogin" />
    </q-card-actions>
  </q-card>
</div>
</div>
</template>

<script>
export default {
  data() {
    return {
      form: {
        username: null,
        password: null,
      },
    };
  },
  methods: {
    handleLogin() {
      console.log("form:", this.form);
    },
  },
};
</script>

```

២. Collection និង Meteor method

ក្នុងមេរៀននេះយើងនឹងប្រើប្រាស់ MongoDB គឺជា Database ។ វានឹងធ្វើការរក្សាទុកនូវទិន្នន័យ (data) ក្នុងទម្រង់ JSON Document នៅក្នុង Collection។

Collection គឺជាកន្លែងសម្រាប់ផ្ទុកនូវ Documents (ទិន្នន័យ) សម្រាប់ប្រើប្រាស់ពេលក្រោយ។

២.១ របៀបបង្កើត collection ក្នុង meteor

ត្រូវបង្កើត file collections.js នៅក្នុង folder

```

-> imports\api\collections
-> imports\api\collections\collections.js

```

រួចធ្វើការសរសេរកូដដើម្បីបង្កើត collection

Example:

```
import { Mongo } from "meteor/mongo";

export const Customer = new Mongo.Collection("customers");
```

២.២ របៀបបង្កើត method ក្នុង meteor

ដំបូងយើងត្រូវបង្កើត file សម្រាប់សរសេរ methods នៅក្នុង folder

-> imports\api\methods

Example: imports\api\methods\customer.js

```
import { Meteor } from "meteor/meteor";
import { Customer } from "../collections/collections";

Meteor.methods({
  "customer.insert"(doc) {
    return Customer.insert(doc);
  },
  "customer.find"() {
    return Customer.find().fetch();
  },
  "customer.update"(doc) {
    return Customer.update({ _id: doc._id }, { $set: doc });
  },
  "customer.remove"(id) {
    return Customer.remove({ _id: id });
  },
});
```

ត្រូវដាក់ទាញចូល file methodទៅដាក់ក្នុង file index.js

-> imports\api\methods\index.js

```
import "../createLink";
import "../customer";
```

២.៣ Collection និង Method សម្រាប់ project

ក.សម្រាប់ Customer

- collection

-> imports\api\collections\collections.js

```
import { Mongo } from "meteor/mongo";
```



```
export const Customer = new Mongo.Collection("customers");
```

- method

-> imports\api\methods\customer.js

```
import { Meteor } from "meteor/meteor";
import { Customer } from "../collections/collections";

Meteor.methods({
  "customer.insert"(doc) {
    return Customer.insert(doc);
  },
  "customer.find"() {
    return Customer.find().fetch();
  },
  "customer.update"(doc) {
    return Customer.update({ _id: doc._id }, { $set: doc });
  },
  "customer.remove"(id) {
    return Customer.remove({ _id: id });
  },
});
```

- import ដាក់ឱ្យស្គាល់នៅលើ server

-> imports\api\methods\index.js

```
import "../createLink";
import "../customer";
```

- ការបោះ method មកលើ

-> imports\ui\components\CustomerForm.vue

```
...
  handleSubmit() {
    this.form.dob = moment(this.form.dob, "YYYY-MM-DD").toDate();
    let method = "customer.insert";
    if (this.updateDoc) {
      method = "customer.update";
    }
    Meteor.call(method, this.form, (err, result) => {
      if (result) {
        this.$emit("close");
      }
    });
  },
  ...
```

-> imports\ui\pages\Customer.vue

```
...
methods: {
  close() {
    this.getData();
    this.updateDoc = null;
    this.dialog = false;
  },
  handleDelete(id) {
    Meteor.call("customer.remove", id, (err, result) => {
      if (result) {
        this.getData();
      }
    });
  },
  handleEdit(doc) {
    this.updateDoc = Object.assign({}, doc);
    this.updateDoc.dob = moment(doc.dob).format("YYYY-MM-DD");

    this.dialog = true;
  },
  getData() {
    Meteor.call("customer.find", (err, result) => {
      if (result) {
        this.customers = result;
      }
    });
  },
},
},
...

```

ខ.សម្រាប់ Category

- Collection

->imports\api\collections\collections.js

```
import { Mongo } from "meteor/mongo";

export const Customer = new Mongo.Collection("customers");
export const Category = new Mongo.Collection("categories");

```

- Method

->imports\api\methods\category.js

```
import { Meteor } from "meteor/meteor";
import { Category } from "../collections/collections";

```

```

Meteor.methods({
  "category.insert"(doc) {
    return Category.insert(doc);
  },
  "category.find"() {
    return Category.find().fetch();
  },
  "category.update"(doc) {
    return Category.update({ _id: doc._id }, { $set: doc });
  },
  "category.remove"(id) {
    return Category.remove({ _id: id });
  },
});

```

- import ដាក់ឱ្យស្គាល់នៅលើ server

->imports\api\methods\index.js

```

import "../createLink";
import "../customer";
import "../category";

```

- ការហៅ method មកប្រើ

->imports\ui\components\CategoryForm.vue

```

...
  handleSubmit() {
    this.form.date = moment(this.form.date, "YYYY-MM-DD").toDate();
    let method = "category.insert";
    if (this.updateDoc) {
      method = "category.update";
    }
    Meteor.call(method, this.form, (err, result) => {
      if (result) {
        this.$emit("close");
      }
    });
  },
...

```

->imports\ui\pages\Category.vue

```

...
  methods: {
    close() {
      this.getData();
      this.updateDoc = null;
      this.dialog = false;
    },
  },

```

```

handleDelete(id) {
  Meteor.call("category.remove", id, (err, result) => {
    if (result) {
      this.getData();
    }
  });
},
handleEdit(doc) {
  this.updateDoc = Object.assign({}, doc);
  this.updateDoc.date = moment(doc.date).format("YYYY-MM-DD");

  this.dialog = true;
},
handleAdd() {
  this.dialog = true;
},
getData() {
  Meteor.call("category.find", (err, result) => {
    if (result) {
      this.categories = result;
    }
  });
},
},
},
...

```

គ.សម្រាប់ Item

- Collection

->imports\api\collections\collections.js

```

import { Mongo } from "meteor/mongo";

export const Customer = new Mongo.Collection("customers");
export const Category = new Mongo.Collection("categories");
export const Item = new Mongo.Collection("items");

```

- Method

->imports\api\methods\item.js

```

import { Meteor } from "meteor/meteor";
import { Item, Category } from "../collections/collections";

Meteor.methods({
  "item.insert"(doc) {
    return Item.insert(doc);
  },
  "item.find"() {
    let data = [];

```

```

    Item.find()
      .fetch()
      .forEach((it) => {
        it.categoryName = Category.findOne({ _id: it.categoryId
      })?.name;
        data.push(it);
      });
    return data;
  },
  "item.update"(doc) {
    return Item.update({ _id: doc._id }, { $set: doc });
  },
  "item.remove"(id) {
    return Item.remove({ _id: id });
  },
});

```

- import ដាក់ឱ្យស្គាល់នៅលើ server

->imports\api\methods\index.js

```

import "../createLink";
import "../customer";
import "../category";
import "../item";

```

- ការបោះ method មកប្រើ

->imports\ui\components\ItemForm.vue

```

...
methods: {
  getCategory() {
    Meteor.call("category.find", (err, result) => {
      if (result) {
        this.categoryOpts = result;
      }
    });
  },

  handleSubmit() {
    this.form.date = moment(this.form.date, "YYYY-MM-DD").toDate();
    let method = "item.insert";
    if (this.updateDoc) {
      method = "item.update";
    }
  }
}

```

```

Meteor.call(method, this.form, (err, result) => {
  if (result) {
    this.$emit("close");
  }
});
},
},
...

```

->imports\ui\pages\Item.vue

```

...
methods: {
  close() {
    this.getData();
    this.updateDoc = null;
    this.dialog = false;
  },
  handleDelete(id) {
    Meteor.call("item.remove", id, (err, result) => {
      if (result) {
        this.getData();
      }
    });
  },
  handleEdit(doc) {
    this.updateDoc = Object.assign({}, doc);
    this.updateDoc.date = moment(doc.date).format("YYYY-MM-DD");
    this.dialog = true;
  },
  handleAdd() {
    this.dialog = true;
  },
  getData() {
    Meteor.call("item.find", (err, result) => {
      if (result) {
        this.items = result;
      }
    });
  },
},
...

```

ឃ.សម្រាប់ Supplier

- Collection

->imports\api\collections\collections.js

```

import { Mongo } from "meteor/mongo";

```

```
export const Customer = new Mongo.Collection("customers");
export const Category = new Mongo.Collection("categories");
export const Item = new Mongo.Collection("items");
export const Supplier = new Mongo.Collection("suppliers");
```

- Method

-> imports\api\methods\supplier.js

```
import { Meteor } from "meteor/meteor";
import { Supplier } from "../collections/collections";

Meteor.methods({
  "supplier.insert"(doc) {
    return Supplier.insert(doc);
  },
  "supplier.find"() {
    return Supplier.find().fetch();
  },
  "supplier.update"(doc) {
    return Supplier.update({ _id: doc._id }, { $set: doc });
  },
  "supplier.remove"(id) {
    return Supplier.remove({ _id: id });
  },
});
```

- import ជាកំហុសស្គាល់នៅលើ server

-> imports\api\methods\index.js

```
import "../createLink";
import "../customer";
import "../category";
import "../item";
import "../supplier";
```

- ការហៅ method មកប្រើ

-> imports\ui\components\SupplierForm.vue

```
...
methods: {
  handleSubmit() {
    let method = this.updateDoc ? `supplier.update` : `supplier.insert`;
    Meteor.call(method, this.form, (error, success) => {
      if (error) {
        console.log("error", error);
      }
      if (success) {
        this.$emit("close");
      }
    });
  }
}
```

```

    }
  });
},
},
...

```

-> imports\ui\pages\Supplier.vue

```

...
methods: {
  close() {
    this.getData();
    this.updateDoc = null;
    this.dialog = false;
  },
  handleDelete(id) {
    Meteor.call("item.remove", id, (error, success) => {
      if (error) {
        console.log("error", error);
      }
      if (success) {
        this.getData();
      }
    });
  },
  handleEdit(doc) {
    console.log("supplier:", doc);
    this.updateDoc = Object.assign({}, doc);
    this.dialog = true;
  },
  handleAdd() {
    this.dialog = true;
  },
  getData() {
    Meteor.call("supplier.find", (error, success) => {
      if (success) {
        this.suppliers = success;
      } else {
        console.log("error", error);
      }
    });
  },
},
...

```

ង.សម្រាប់ Purchase

- Collection

->imports\api\collections\collections.js


```
import { Mongo } from "meteor/mongo";

export const Customer = new Mongo.Collection("customers");
export const Category = new Mongo.Collection("categories");
export const Item = new Mongo.Collection("items");
export const Supplier = new Mongo.Collection("suppliers");
export const Purchase = new Mongo.Collection("purchases");
export const Import = new Mongo.Collection("imports");
```

- Method

->imports\api\methods\purchase.js

```
import { Meteor } from "meteor/meteor";
import { Purchase, Import, Item } from "../collections/collections";

Meteor.methods({
  "purchase.add"(doc) {
    console.log("doc", doc);
    let importDoc = Object.assign({}, doc);
    let oldPurchase = Purchase.findOne({ itemId: doc.itemId });
    if (!oldPurchase) {
      importDoc.remains = 0;
      Import.insert(importDoc);
      return Purchase.insert(doc);
    } else {
      importDoc.remains = oldPurchase.qty;
      Import.insert(importDoc);
      oldPurchase.qty += doc.qty;
      oldPurchase.cost = doc.cost;
      oldPurchase.price = doc.price;
      oldPurchase.date = doc.date;
      return Purchase.update({ _id: oldPurchase._id }, { $set: oldPurchase });
    }
  },
  "purchase.find"() {
    let data = [];
    Purchase.find().forEach((doc) => {
      let item = doc;
      item.name = Item.findOne({ _id: doc.itemId })?.name;
      data.push(item);
    });
    return data;
  },
});
```

- import ដាក់ឱ្យស្ថាននៅលើ server

->imports\api\methods\index.js

```
import "../createLink";
```

```
import "../customer";
import "../category";
import "../item";
import "../supplier";
import "../purchase";
```

• ការបោះ method មកប្រើ

-> imports\ui\components\PurchaseForm.vue

```
...
methods: {
  handleSubmit() {
    let valid = false;
    for (const doc in this.form) {
      if (this.form[doc] == null) {
        valid = true;
      }
    }
    if (!valid) {
      this.form.date = moment(this.form.date, "YYYY-MM-DD").toDate();
      Meteor.call("purchase.add", this.form, (err, result) => {
        if (result) {
          this.$emit("close");
        } else {
          console.log("err", err);
        }
      });
    } else {
      Notify.create({
        message: "Please fill the form ",
        color: "negative",
        type: "negative",
        position: "top",
      });
    }
  },
  getItem() {
    Meteor.call("item.find", (err, result) => {
      if (result) {
        this.itemOpts = result;
      }
    });
  },
  getSupplier() {
    Meteor.call("supplier.find", (err, result) => {
      if (result) {
        console.log("result", result);
        this.supplierOpts = result;
      } else {
        console.log("err", err);
      }
    });
  },
}
```

```

    }
  });
},
},
...

```

-> imports\ui\pages\Purchase.vue

```

...
methods: {
  handleAdd() {
    this.dialog = true;
  },
  close() {
    this.getData();
    this.dialog = false;
  },
  getData() {
    Meteor.call("purchase.find", (error, result) => {
      if (result) {
        this.items = result;
      } else {
        console.log("err", error);
      }
    });
  },
},
},
...

```

២.សម្រាប់ Sale

- Collection Invoice

->imports\api\collections\collections.js

```

import { Mongo } from "meteor/mongo";

export const Customer = new Mongo.Collection("customers");
export const Category = new Mongo.Collection("categories");
export const Item = new Mongo.Collection("items");
export const Supplier = new Mongo.Collection("suppliers");
export const Purchase = new Mongo.Collection("purchases");
export const Import = new Mongo.Collection("imports");
export const Export = new Mongo.Collection("exports");

```

- Method

->imports\api\methods\invoice.js

```

import { Meteor } from "meteor/meteor";
import { Purchase, Export } from "../collections/collections";

```

```
Meteor.methods({
  "invoice.number"() {
    return Export.find().count() + 1;
  },
  "invoice.insert"(doc) {
    let soldItems = doc.items;
    Purchase.find().forEach((doc) => {
      soldItems.forEach((soldItem) => {
        if (soldItem._id === doc.itemId) {
          let remainQty = doc.qty - soldItem.qty;
          if (remainQty === 0) {
            Purchase.remove({ _id: doc._id });
          } else {
            Purchase.update({ _id: doc._id }, { $set: { qty: remainQty } });
          }
        }
      });
    });
    return Export.insert(doc);
  },
});
```

- import ដាក់ប្រស្តូរនៅលើ server

->imports\api\methods\index.js

```
import "../createLink";
import "../customer";
import "../category";
import "../item";
import "../inventory";
import "../invoice";
```

- ការបោះ method មកលើ

->imports\ui\pages\Order.vue

```
...
methods: {
  handleSubmit() {
    let doc = {
      invoiceNumber: this.form.invoiceNumber,
      customerId: this.form.customerId,
      date: moment(this.form.date, "YYYY-MM-DD").toDate(),
      items: [],
    };
    this.form.items.forEach((it) => {
      doc.items.push({
        _id: it._id,
        price: it.price,
        qty: it.qty,
```

```

    });
  });

  Meteor.call("invoice.insert", doc, (err, result) => {
    if (result) {
      this.clearForm();
    }
  });
},
handleChangeItem(val) {
  let found = this.form.items.findIndex((doc) => {
    return doc._id == val._id;
  });
  if (found != -1) {
    this.form.items[found].qty++;
  } else {
    this.form.items.push({
      _id: val.itemId,
      name: val.name,
      price: val.price,
      qty: 1,
    });
  }
},
handleRemove(index) {
  this.$delete(this.form.items, index);
},
total() {
  let total = 0;
  this.form.items.forEach((it) => {
    total += it.price * it.qty;
  });
  return total;
},
clearForm() {
  this.form = {
    invoiceNumber: null,
    date: moment(new Date()).format("YYYY-MM-DD"),
    customerId: null,
    items: [],
  };
  this.item = null;
  this.getInvoiceNumber();
},
getInvoiceNumber() {
  Meteor.call("invoice.number", (err, result) => {
    if (result) {
      this.form.invoiceNumber = result;
    }
  });
});

```

```

    },
    getPurchase() {
      Meteor.call("purchase.find", (err, result) => {
        if (result) {
          this.itemOpts = result;
        }
      });
    },
    getCustomer() {
      Meteor.call("customer.find", (err, result) => {
        if (result) {
          this.customerOpts = result;
        }
      });
    },
  },
},
...

```

៣.សម្រាប់ Import

- Method

->imports\api\methods\import.js

```

import { Meteor } from "meteor/meteor";
import moment from "moment";
import { Import, Item, Supplier } from "../collections/collections";

Meteor.methods({
  "import.findByDate"(date) {
    let data = [];
    let fromDate = moment(date.fromDate, "DD-MM-YYYY");
    let toDate = moment(date.toDate, "DD-MM-YYYY");
    Import.find().forEach((doc) => {
      let item = doc;
      let date = moment(doc.date, "DD-MM-YYYY");
      if (date >= fromDate && date <= toDate) {
        item.name = Item.findOne({ _id: doc.itemId })?.name;
        item.company = Supplier.findOne({ _id: doc.supplierId })?.company;
        data.push(item);
      }
    });
    return data;
  },
});

```

- import ដាក់ឱ្យស្គាល់នៅលើ server

->imports\api\methods\index.js

...

```
import "../import";
```

- ការហៅ method មកប្រើ

```
->imports\ui\pages\Import.vue
```

```
...
searchByDate(date) {
  Meteor.call("import.findByDate", date, (err, result) => {
    if (result) {
      this.items = result;
    }
  });
},
...

```

៨.សម្រាប់ Export

- Add Method

```
->imports\api\methods\invoice.js
```

```
import { Meteor } from "meteor/meteor";
import moment from "moment";
import { Export, Purchase, Item, Customer } from "../collections/collections";

Meteor.methods({
  "invoice.number"() {
    return Export.find().count() + 1;
  },
  "invoice.insert"(doc) {
    let soldItems = doc.items;
    Purchase.find().forEach((doc) => {
      soldItems.forEach((soldItem) => {
        if (soldItem._id === doc.item_id) {
          let remainQty = doc.qty - soldItem.qty;
          if (remainQty === 0) {
            Purchase.remove({ _id: doc._id });
          } else {
            Purchase.update({ _id: doc._id }, { $set: { qty: remainQty } });
          }
        }
      });
    });
    return Export.insert(doc);
  },
  "invoice.find"(date) {
    let data = [];
    let fromDate = moment(date.fromDate, "DD-MM-YYYY");
    let toDate = moment(date.toDate, "DD-MM-YYYY");
  }
});

```

```
Export.find().forEach((doc) => {
  let date = moment(doc.date, "DD-MM-YYYY");
  if (date >= fromDate && date <= toDate) {
    let newDoc = doc;
    newDoc.customerName = Customer.findOne({ _id: doc.customerId
  })?.name;
    let items = [];
    doc.items.forEach((it) => {
      it.name = Item.findOne({ _id: it._id })?.name;
      items.push(it);
    });
    newDoc.items = items;
    data.push(newDoc);
  }
});
console.log("data:", data);
return data;
},
});
```

• ការហៅ method មកប្រើ

-> imports\ui\pages\Export.vue

```
...
methods: {
  searchByDate(date) {
    Meteor.call("invoice.find", date, (err, result) => {
      if (result) {
        this.items = result;
      }
    });
  },
},
},
...

```

២.៣ Printjs

ការដំឡើង printjs package

```
npm install print-js --save
```

Import the library into your project:

```
import printJS from 'print-js'
```

ឧទាហរណ៍

-> imports\ui\pages\Import.vue


```

<template>
  <div>
    <form-search-by-date @searchByDate="searchByDate" />
    <div class="q-mt-md">
      <q-btn icon="print" @click="handlePrint" />
      <q-markup-table id="print" :dense="dense">
        <thead>
          <tr>
            <th class="text-left">No</th>
            <th class="text-left">Date</th>
            <th class="text-left">Company</th>
            <th class="text-left">Item</th>
            <th class="text-left">Remains</th>
            <th class="text-left">Cost</th>
            <th class="text-left">Price</th>
            <th class="text-left">Quantity</th>
            <th class="text-left">Status</th>
          </tr>
        </thead>
        <tbody>
          <tr v-for="(item, index) in items" :key="index">
            <td class="text-left">{{ index + 1 }}</td>
            <td class="text-left">{{ item.date | toDate }}</td>
            <td class="text-left">{{ item.company }}</td>
            <td class="text-left">{{ item.name }}</td>
            <td class="text-left">{{ item.remains }}</td>
            <td class="text-left">{{ item.cost }}</td>
            <td class="text-left">{{ item.price }}</td>
            <td class="text-left">{{ item.qty }}</td>
            <td class="text-left">{{ item.status }}</td>
          </tr>
        </tbody>
      </q-markup-table>
    </div>
  </div>
</template>

<script>
import moment from "moment";
import FormSearchByDate from "../components/FormSearchByDate.vue";
import printJS from "print-js";
export default {
  components: { FormSearchByDate },
  data() {
    return {
      items: [],
      dense: false,
    };
  },
  filters: {

```

```

    toDate(date) {
      return moment(date).format("DD-MM-YYYY");
    },
  },
  methods: {
    searchByDate(date) {
      console.log("date", date);
      Meteor.call("import.findByDate", date, (err, result) => {
        if (result) {
          this.items = result;
        }
      });
    },
    handlePrint() {
      this.dense = true;
      setTimeout(() => {
        printJS({
          printable: "print",
          type: "html",
          targetStyles: ["*"],
        });
        this.dense = false;
      });
    },
  },
};
</script>

```

ក. Print Invoice

- Create Component

imports\ui\components\PrintInvoice.vue

```

<template>
  <div id="print">
    <q-card>
      <q-card-section>
        <div class="row">
          <div class="col text-center">
            <div class="text-h6">Master IT-Shop</div>
            <div class="text-h6">សង្កាត់រតនៈ ស្រុកបាត់ដំបង ខេត្តបាត់ដំបង</div>
            <div class="text-h6">
              លក់ទំនិញគ្រប់ប្រភេទ 017 292 912/069 292 912
            </div>
            <div class="text-h6">
              អតិថិជន: {{ data.customer }} | ថ្ងៃទី: {{ data.date }} |
              លេខវិក័យប័ត្រ: {{ data.invoiceNumber }}
            </div>
          </div>
        </div>
      </div>
    </div>
  </div>
</template>

```

```

</q-card-section>
<q-markup-table dense>
  <thead>
    <tr>
      <th class="text-left">No</th>
      <th class="text-left">Item</th>
      <th class="text-left">Quantity</th>
      <th class="text-left">Price</th>
      <th class="text-left">Amount</th>
    </tr>
  </thead>
  <tbody>
    <tr v-for="(item, index) in data.items" :key="index">
      <td>{{ index + 1 }}</td>
      <td>{{ item.name }}</td>
      <td>{{ item.qty }}</td>
      <td>{{ item.price }}</td>
      <td>{{ item.qty * item.price }}</td>
    </tr>
    <tr>
      <td colspan="4">Total</td>
      <td colspan="1">{{ total() }}</td>
    </tr>
  </tbody>
</q-markup-table>
</q-card>
</div>
</template>
<script>
import printJS from "print-js";
export default {
  props: {
    data: {
      required: true,
      type: Object,
    },
  },
  mounted() {
    console.log("data", this.data);
    printJS({
      printable: "print",
      type: "html",
      targetStyles: ["*"],
    });
    this.$emit("close");
  },
  methods: {
    total() {
      let amount = 0;
      this.data.items.forEach((doc) => {

```

```

        amount += doc.price * doc.qty;
    });
    return amount;
  },
};
};
</script>

```

- Usage

```
imports\ui\pages\Sale.vue
```

```

<template>
  <div class="q-pt-md q-mr-xl q-ml-xl">
    <q-dialog
      v-model="dialog"
      persistent
      :maximized="true"
      transition-show="slide-up"
      transition-hide="slide-down"
    >
      <q-card>
        <q-bar>
          <q-space />
          <q-btn
            dense
            flat
            icon="close"
            color="white"
            class="bg-red shadow-4"
            rounded
            v-close-popup
          >
            <q-tooltip content-class="bg-white text-primary ">Close</q-
tooltip>
          </q-btn>
        </q-bar>
        <print-invoice :data="form" @close="clearForm" />
      </q-card>
    </q-dialog>
    <form @submit.prevent="handleSubmit">
      <q-card class="shadow-6">
        <q-card-section class="text-h5 text-center">Sale Form </q-card-
section>
        <q-card-section>
          <div class="row items-center">
            <div class="col-md-8 col-xs-12">
              <div class="col-xs-12 col-sm-4 q-ma-sm">
                <q-input
                  square
                  outlined
                  type="number"

```

```

        readonly
        v-model.number="form.invoiceNumber"
        label="Invoice Number"
      />
    </div>
    <div class="col-xs-12 col-sm-4 q-ma-sm">
      <q-input
        square
        outlined
        readonly
        type="date"
        v-model="form.date"
        label="Date"
      />
    </div>
    <div class="col-xs-12 col-sm-4 q-ma-sm">
      <q-select
        square
        outlined
        v-model="form.customerId"
        :options="customerOpts"
        label="Customer"
        map-options
        emit-value
        option-label="name"
        option-value="_id"
        required
      />
    </div>
    <div class="col-xs-12 col-sm-4 q-ma-sm">
      <q-select
        square
        outlined
        v-model="item"
        :options="itemOpts"
        label="Items"
        emit-value
        map-options
        option-label="name"
        @input="handleChangeItem"
      />
    </div>
  </div>
  <div class="col-md-4 col-xs-12">
    <q-card flat class="q-pa-xl q-pr-xl">
      <div class="text-h5 text-center">Total : {{ total() }}</div>
    </q-card>
  </div>
</div>

```

```

<q-markup-table>
  <thead>
    <tr>
      <th class="text-left">No</th>
      <th class="text-left">Name</th>
      <th class="text-left">Price</th>
      <th class="text-left">Qty</th>
      <th class="text-left">Amount</th>
      <th class="text-left">Action</th>
    </tr>
  </thead>
  <tbody>
    <tr v-for="(item, index) in form.items" :key="index">
      <td class="text-left">{{ index + 1 }}</td>
      <td class="text-left">{{ item.name }}</td>
      <td class="text-left">{{ item.price }}</td>
      <td class="text-left" style="width: 150px">
        <q-input type="number" v-model.number="item.qty" />
      </td>
      <td class="text-left">{{ item.price * item.qty }}</td>
      <td class="text-left">
        <q-btn
          color="red"
          icon="delete"
          dense
          flat
          @click="handleRemove(index)"
        />
      </td>
    </tr>
  </tbody>
</q-markup-table>
<div
  class="text-center q-mt-lg"
  v-if="form.items.length != 0 && form.customerId"
>
  <q-btn color="primary" label="Submit" type="submit" />
</div>
</q-card-section>
</q-card>
</form>
</div>
</template>

<script>
import moment from "moment";
import PrintInvoice from "../components/PrintInvoice.vue";
export default {
  components: { PrintInvoice },
  data() {

```

```

return {
  item: null,
  form: {
    invoiceNumber: null,
    date: moment(new Date()).format("YYYY-MM-DD"),
    customerId: null,
    items: [],
  },
  itemOpts: [],
  customerOpts: [],
  dialog: false,
};
},
mounted() {
  this.getInvoiceNumber();
  this.getCustomer();
  this.getPurchase();
},
methods: {
  handleSubmit() {
    let doc = {
      invoiceNumber: this.form.invoiceNumber,
      customerId: this.form.customerId,
      date: moment(this.form.date, "YYYY-MM-DD").toDate(),
      items: [],
    };
    this.form.items.forEach((it) => {
      doc.items.push({
        _id: it._id,
        price: it.price,
        qty: it.qty,
      });
    });
    let index = this.customerOpts.findIndex((cus) => {
      return cus._id == this.form.customerId;
    });
    this.form.customer = this.customerOpts[index].name;
    Meteor.call("invoice.insert", doc, (err, result) => {
      if (result) {
        this.dialog = true;
      }
    });
  },
  handleChangeItem(val) {
    let found = this.form.items.findIndex((doc) => {
      return doc._id == val._id;
    });
    if (found != -1) {
      this.form.items[found].qty++;
    } else {

```

```

        this.form.items.push({
          _id: val.itemId,
          name: val.name,
          price: val.price,
          qty: 1,
        });
      },
    },
    handleRemove(index) {
      this.$delete(this.form.items, index);
    },
    total() {
      let total = 0;
      this.form.items.forEach((it) => {
        total += it.price * it.qty;
      });
      return total;
    },
    clearForm() {
      this.dialog = false;
      this.form = {
        invoiceNumber: null,
        date: moment(new Date()).format("YYYY-MM-DD"),
        customerId: null,
        items: [],
      };
      this.item = null;
      this.getInvoiceNumber();
    },
    getInvoiceNumber() {
      Meteor.call("invoice.number", (err, result) => {
        if (result) {
          this.form.invoiceNumber = result;
        }
      });
    },
    getPurchase() {
      Meteor.call("purchase.find", (err, result) => {
        if (result) {
          this.itemOpts = result;
        }
      });
    },
    getCustomer() {
      Meteor.call("customer.find", (err, result) => {
        if (result) {
          this.customerOpts = result;
        }
      });
    },
  },
},

```



```
};
</script>
```

២. User Authentication

- User Backend

ដំឡើង accounts-password

```
meteor add accounts-password
```

ការបង្កើត methods សម្រាប់ user

->imports/api/methods/user.js

```
import { Meteor } from "meteor/meteor";
import { Accounts } from "meteor/accounts-base";
Meteor.methods({
  "user.create"(doc) {
    return Accounts.createUser({
      username: doc.username,
      password: doc.password,
      email: doc.email,
      profile: {
        phone: doc.phone,
        role: doc.role,
      },
    });
  },
  "user.update"(doc) {
    if (doc.reset) {
      Accounts.setPassword(doc._id, doc.password);
    }
    let emails = Meteor.users.findOne({ _id: doc._id }).emails;
    emails[0].address = doc.email;
    return Meteor.users.update(
      { _id: doc._id },
      {
        $set: {
          username: doc.username,
          emails: emails,
          profile: {
            phone: doc.phone,
            role: doc.role,
          },
        },
      },
    );
  },
  "user.find"() {
```

```
let data = [];
Meteor.users
  .find()
  .fetch()
  .forEach((it) => {
    let doc = {
      _id: it._id,
      username: it.username,
      email: it.emails[0].address,
      phone: it.profile.phone,
      role: it.profile.role,
    };
    data.push(doc);
  });
return data;
},
"user.remove"(id) {
  return Meteor.users.remove({ _id: id });
},
});
```

- import method ដាក់ឱ្យស្គាល់នៅលើ server

-> imports\api\methods\index.js

```
...
• import "./user";
```

- ការប្រើប្រាស់ Method

-> imports/ui/pages/User.vue

```
...
getData() {
  Meteor.call("user.find", (error, result) => {
    if (error) {
      console.log("error", error);
    }
    if (result) {
      this.users = result;
    }
  });
},
handleDelete(id) {
  Meteor.call("user.remove", id, (err, result) => {
    if (result) {
      this.getData();
    }
  });
},
...

```

->imports\ui\components\UserForm.vue

```

...
handleSubmit() {
  let method = "user.create";
  if (this.form.password == this.confirm && this.form.password != null) {
    if (this.reset) {
      method = "user.update";
      this.form.reset = this.reset;
    }
  }
  if (this.updateDoc) {
    method = "user.update";
  }
  Meteor.call(method, this.form, (error, result) => {
    if (result) {
      this.$emit("close");
    } else {
      console.log("err:", error);
    }
  });
},
...

```

៥. Start Up

-> imports\api\methods\index.js

```

import "./createLink";
import "./customer";
import "./category";
import "./item";
import "./supplier";
import "./purchase";
import "./invoice";
import "./import";
import "./user";
import { Meteor } from "meteor/meteor";

Meteor.startup(() => {
  if (Meteor.users.find().count() == 0) {
    let user = {
      username: "admin",
      password: "admin",
      email: "admin@admin.com",
      phone: "017292912",
      role: "admin",
    };
    Meteor.call("user.create", user, (err, result) => {
      if (result) {
        console.log("created user:", result);
      }
    });
  }
});

```

```

    }
  });
}
});

```

៥.Login

- ការកំណត់ authentication

-> imports\ui\routes\index.js

```

import Vue from "vue";
import VueRouter from "vue-router";
Vue.use(VueRouter);
let routes = [
  {
    path: "/",
    name: "Home",
    component: () => import("../pages/Home.vue"),
    meta: {
      requiresAuth: true,
    },
  },
  {
    path: "/user",
    name: "User",
    component: () => import("../pages/User.vue"),
    meta: {
      requiresAuth: true,
    },
  },
  {
    path: "/customer",
    name: "Customer",
    component: () => import("../pages/Customer.vue"),
    meta: {
      requiresAuth: true,
    },
  },
  {
    path: "/category",
    name: "Category",
    component: () => import("../pages/Category.vue"),
    meta: {
      requiresAuth: true,
    },
  },
  {
    path: "/item",

```

```

    name: "Item",
    component: () => import("../pages/Item.vue"),
    meta: {
      requiresAuth: true,
    },
  },
  {
    path: "/supplier",
    name: "Supplier",
    component: () => import("../pages/Supplier.vue"),
    meta: {
      requiresAuth: true,
    },
  },
  {
    path: "/purchase",
    name: "Purchase",
    component: () => import("../pages/Purchase.vue"),
    meta: {
      requiresAuth: true,
    },
  },
  {
    path: "/sale",
    component: () => import("../pages/Sale.vue"),
    meta: {
      requiresAuth: true,
    },
  },
  {
    path: "/import",
    name: "Import",
    component: () => import("../pages/Import.vue"),
    meta: {
      requiresAuth: true,
    },
  },
  {
    path: "/export",
    name: "Export",
    component: () => import("../pages/Export.vue"),
    meta: {
      requiresAuth: true,
    },
  },
  {
    path: "/login",
    component: () => import("../pages/Login.vue"),
  },
];

```

```
const router = new VueRouter({
  routes,
  mode: "history",
});
router.beforeEach((to, from, next) => {
  if (to.meta.requiresAuth) {
    if (!Meteor.userId()) {
      next({
        path: "/login",
      });
    } else {
      next();
    }
  } else {
    next();
  }
});
export default router;
```

- ការប្រើប្រាស់ Login method

```
imports\ui\pages\Login.vue
```

```
...
handleLogin() {
  Meteor.loginWithPassword(
    this.form.username,
    this.form.password,
    (err) => {
      if (!err) {
        console.log("user:", Meteor.user());
        this.$router.push("/");
      } else {
      }
    }
  );
},
...

```

៥.Layout

បង្កើត folder layouts

```
imports/ui/layouts
```

បង្កើត Layout ចំនួន២គឺ

```
->imports/ui/layouts/AdminLayout.vue
```

Copy from App.vue

```
<template>
```

```

<div>
  <q-layout view="hHr LpR lfr">
    <q-header class="bg-primary text-white" elevated>
      <q-toolbar>
        <q-
btn flat @click="drawer = !drawer" round dense icon="menu" />
        <q-toolbar-title>Master IT Shop</q-toolbar-title>
      </q-toolbar>
    </q-header>

    <q-drawer v-model="drawer" content-class="bg-grey-3" show-if-
above>
      <q-scroll-area class="fit">
        <q-list class="menu">
          <q-item class="item" clickable v-ripple exact to="/">
            <q-item-section avatar>
              <q-icon name="home" />
            </q-item-section>
            <q-item-section> Home </q-item-section>
          </q-item>
          <q-item class="item" clickable v-ripple exact to="/user">
            <q-item-section avatar>
              <q-icon name="person" />
            </q-item-section>
            <q-item-section> User </q-item-section>
          </q-item>
          <q-item class="item" clickable v-ripple to="/customer">
            <q-item-section avatar>
              <q-icon name="groups" />
            </q-item-section>
            <q-item-section> Customer </q-item-section>
          </q-item>
          <q-item class="item" clickable v-ripple to="/category">
            <q-item-section avatar>
              <q-icon name="backup_table" />
            </q-item-section>
            <q-item-section> Category </q-item-section>
          </q-item>
          <q-item class="item" clickable v-ripple to="/item">
            <q-item-section avatar>
              <q-icon name="dns" />
            </q-item-section>
            <q-item-section> Item </q-item-section>
          </q-item>
          <q-item class="item" clickable v-ripple to="/supplier">
            <q-item-section avatar>
              <q-icon name="maps_home_work" />
            </q-item-section>
            <q-item-section> Supplier </q-item-section>
          </q-item>
        </q-list>
      </q-scroll-area>
    </q-drawer>
  </q-layout>
</div>

```

```

    <q-item class="item" clickable v-ripple to="/purchase">
      <q-item-section avatar>
        <q-icon name="inventory" />
      </q-item-section>
      <q-item-section> Purchase </q-item-section>
    </q-item>
    <q-separator />
    <q-item class="item" clickable v-ripple to="/sale">
      <q-item-section avatar>
        <q-icon name="chrome_reader_mode" />
      </q-item-section>
      <q-item-section> Sale </q-item-section>
    </q-item>
    <q-list bordered class="item">
      <q-expansion-item icon="report" label="Report">
        <q-item class="item" clickable v-ripple to="import">
          <q-item-section avatar>
            <q-icon name="arrow_circle_down" />
          </q-item-section>
          <q-item-section> Import </q-item-section>
        </q-item>
        <q-item class="item" clickable v-ripple to="export">
          <q-item-section avatar>
            <q-icon name="arrow_circle_up" />
          </q-item-section>
          <q-item-section> Export </q-item-section>
        </q-item>
      </q-expansion-item>
    </q-list>
    <q-item class="item" clickable v-
ripple @click="handleLogout">
      <q-item-section avatar>
        <q-icon name="logout" />
      </q-item-section>
      <q-item-section> Logout </q-item-section>
    </q-item>
  </q-list>
</q-scroll-area>
</q-drawer>

<q-page-container>
  <q-page class="q-mt-md q-ml-md q-mr-md">
    <div class="text-center text-h5">{{ $route.name }}</div>
    <router-view />
  </q-page>
</q-page-container>
</q-layout>
</div>
</template>
<script>
export default {

```



```

data() {
  return {
    drawer: false,
  };
},
methods: {
  handleLogout() {
    Meteor.logout(() => {
      this.$router.push("/login");
    });
  },
},
};
</script>
<style>
.menu :hover.item {
  transform: scale(1.15);
  transition: transform 0.2s;
  background: rgb(245, 245, 245);
  margin-left: 20px;
  color: rgb(1, 98, 255);
}
.menu .item {
  transform: scale(1);
  transition: transform 1s;
}
</style>

```

->imports/ui/layouts/PublicLayout.vue

```

<template>
  <router-view />
</template>

```

ធ្វើការកែប្រែលើ File App.vue

->imports/ui/App.vue

```

<template>
  <component :is="currentLayout" />
</template>
<script>
import AdminLayout from "../layouts/AdminLayout.vue";
import PublicLayout from "../layouts/PublicLayout.vue";
export default {
  components: {
    AdminLayout,
    PublicLayout,
  },
  data() {

```

```
return {
  currentLayout: null,
};
},
watch: {
  $route: {
    immediate: true,
    handler(newValue, oldValue) {
      if (newValue.meta.requiresAuth && newValue.name !== "Home") {
        this.currentLayout = "AdminLayout";
      } else {
        this.currentLayout = "PublicLayout";
      }
    },
  },
},
},
};
</script>
```

ច.ការដំឡើង

- Meteor Production

meteor build PATH

example: meteor build ~/Desktop/production

- Install Node
- Install MongoDB

ធ្វើការដោតឡូត [Mongodb Service](#) រួចធ្វើការដំឡើង

- Set variable
- MONGO_URL
SETX MONGO_URL mongodb://localhost:27017/myapp
- PORT
SETX PORT 3000
- ROOT_URL
- SETX ROOT_URL http://localhost:3000
- [Fixing Node-Gyp](#)
- Install [Python](#)
- Install Visual C++ Build Environment: [Visual Studio Build Tools](#)
- Install "Desktop development with C++"
- Extraction and run

ចូលទៅកាន់ទីតាំង PATH ដែល file production តាំងនៅ។

example: `cd ~/Desktop/production`

រួចធ្វើការពន្លាត file production

example: `tar -xvzf lesson-project.tar.gz`

រួចចូលទៅកាន់ folder bundle

Example: `cd bundle`

ធ្វើការ Install package

`cd programs/server`

`npm i`

រួចចូលទៅកាន់ folder bundle វិញរួចធ្វើការ Run `node main.js`

Open browser enter: <http://localhost:3000/>

